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Generative AI and Legal Aid: Results from a Field Study and 100 Use Cases to Bridge the Access to Justice Gap

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GENERATIVE AI AND LEGAL AID: RESULTS FROM A FIELD STUDY AND 100 USE CASES TO BRIDGE THE ACCESS TO JUSTICE GAP

*Colleen V. Chien and Miriam Kim**

How can AI tools be used to address the access to justice gap—the 90% of low-income Americans that lack adequate legal assistance? We conducted the first field study of lawyers using generative AI of which we are aware and a companion survey of 202 legal aid professionals to find out. A cohort of ninety-one people received up to two months of access to paid generative artificial intelligence tools, a randomly selected subset of which also received "concierge" support, including peer use cases, office hours, and assistance. Following the pilot, 90% of pilot participants reported increased productivity, and 75% reported their intent to continue using generative AI tools. While concerns remained, pilot participants managed risks by focusing on lower-risk applications like document summarization, confirmatory or preliminary research, the production of first drafts, and translation, from legalese or English into more accessible formats. Before the trial, women were far less likely than men to use or value the tools. By the trial's end, men's and women's outcomes across various measures were statistically indistinguishable. Participants receiving concierge services had significantly better outcomes than their control group counterparts across a range of metrics.

These results suggest that generative AI tools can significantly enhance the provision of legal aid services, but that how they are introduced matter—though women comprise the majority of public interest

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lawyers, organic uptake of generative AI tools was much higher among men in our study. Assistance can also improve tool adoption. The participants' positive experiences support viewing AI technologies as augmenting rather than threatening the work of lawyers. In contrast to equipping lawyers with generative AI tools, legal-aid lawyer-directed technological solutions may have the greatest potential to not just marginally, but dramatically, increase service coverage. We suggest some steps, such as exploring regulatory sandboxes and devising ways to institute voluntary certification or "seal of approval" programs verifying the quality of legal aid bots to support such generative collaborations. Along with the paper, we release a companion database of one hundred helpful use cases, including prompts and outputs, provided by legal aid professionals in the trial, to support broader adoption of AI tools.

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INTRODUCTION

While AI technologies are increasing productivity across a wide variety of domains, their distributional impacts remain uncertain.[1] This study considers the potential of AI tools to address the access to justice gap, the high proportion of low-income Americans with unmet legal needs, through the augmentation of legal aid workers. While AI introduces new possibilities to expand the reach of justice—for example, by increasing the accessibility of legal logic, processes, and filings across income, language, and geographical barriers[2]—obstacles remain. First, generative AI tools hallucinate, limiting their use in high-precision contexts like the law.[3] Second, resource disparities mean that specialized tools being developed to overcome the shortcomings of general purpose products will not necessarily be accessible to

1. At the macro level, much of the focus has been on the impact of AI on jobs. Although automation generally disfavors workers with low skills and education, fueling estimates of worsening inequality in sectors like transportation, as drivers are replaced by autonomous vehicles, generative AI's ability to automate knowledge work has been associated with an equalizing effect across workers as discussed in Part I. *Compare The Philosophy of AI: Learning from History, Shaping Our Future: Hearing Before the S. Comm. on Homeland Sec. & Gov't Affs.*, 117th Cong. (2023) (statement of Daron Acemoglu, Institute Professor, MIT), https://www.hsgac.senate.gov/wp-content/uploads/Testimony-Acemoglu-2023-11-08.pdf [https://perma.cc/36YG-NNAZ] (providing an overview of the potential role of AI and automation to widen inequality and the importance of policy and other non-technical factors in shaping their impact), *with* RAKESH KOCCHAR, PEW RSCH. CTR., WHICH U.S. WORKERS ARE MORE EXPOSED TO AI ON THEIR JOBS? 6–7 (2023) https://www.pewresearch.org/social-trends/wp-content/uploads/sites/3/2023/07/st_2023.07.26_ai-and-jobs.pdf [https://perma.cc/UZ2P-3KFQ] (finding that jobs that require a college education and analytical skills are more likely to be vulnerable to AI displacement); *see also* ALEXANDRE GEORGIEFF & RAPHAELA HYEE, ORG. FOR ECON. COOP. & DEV., ARTIFICIAL INTELLIGENCE AND EMPLOYMENT: NEW CROSS-COUNTRY EVIDENCE 3, 36–48 (2021), https://www.oecd-ilibrary.org/social-issues-migration-health/artificial-intelligence-and-employment_c2c1d276-en.pdf [http://perma.cc/B6RW-WH6B] (finding that exposure to AI is positively associated with employment growth in occupations where computer use is high, such as managers and professionals, but negatively associated with growth in working hours in occupations where computer use is low, such as agricultural workers and machine operators); ANTON KORINEK ET AL., INT'L MONETARY FUND, TECHNOLOGICAL PROGRESS, ARTIFICIAL INTELLIGENCE, AND INCLUSIVE GROWTH (2021), https://www.imf.org/-/media/Files/Publications/WP/2021/English/wpiea2021166-print-pdf.ashx [https://perma.cc/CML3-PQ82] (discussing mechanisms by which technological progress can amplify or reduce economic inequality); Michael Webb, The Impact of Artificial Intelligence on the Labor Market 1 (Jan. 11, 2019) (unpublished manuscript), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3482150 [https://perma.cc/U3XD-2EX3] (characterizing AI as directed towards high-skilled tasks and predicting that AI will reduce top decile, but not top 1%, inequality); VALERIO CAPRARO ET AL., THE IMPACT OF GENERATIVE ARTIFICIAL INTELLIGENCE ON SOCIOECONOMIC INEQUALITIES AND POLICY MAKING (Dec. 16, 2023), https://arxiv.org/pdf/2401.05377.pdf [https://perma.cc/V5TN-7AK5] (identifying, across four domains—work, education, health, and information—ways in which generative AI could worsen existing inequalities but also resolve persistent social problems).

2. *See* discussion *infra* Part I.

3. *See* discussion *infra* Part IV.

resource-limited lawyers and consumers.[4] Third, despite the notion that everyone benefits from innovation as it trickles down,[5] there is abundant evidence that technology often fails to be a leveler[6] and that low-income and marginalized individuals often do not reap the full benefits of product innovation—because it is not priced or otherwise within reach, deprioritizes their unique needs,[7] or tends to cater to the better off.[8]

A major channel of legal assistance to low-income consumers is through state and philanthropically funded legal aid organizations, which represent individuals in consequential matters such as eviction defense, deportation proceedings, and criminal defense.[9] Legal aid services are often oversubscribed, addressing only a fraction of the need, implying that increasing the productivity of legal aid professionals could appreciably increase access to justice.[10]

To explore the potential of generative AI tools to increase access to justice by augmenting the work of legal aid workers, we surveyed

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4. Bob Ambrogi, *12 Thoughts on Promises & Challenges of AI in Legal After Yesterday's AI Summit at Harvard Law School*, LAWSITES (Sept. 23, 2023), https://www.lawnext.com/2023/09/thoughts-on-promises-and-challenges-of-ai-in-legal-after-yesterdays-ai-summit-at-harvard-law-school.html [https://perma.cc/YVQ2-UKWN]. At least not initially, see profile of Co-Counsel *infra* at Part I, and Harvey.ai, which were first piloted among high-end law firms. Charlotte Johnstone, *Macfarlanes Showed Us How It's Using Harvey AI. Here's What We Learnt*, LAW.COM (Oct. 9, 2023, 5:38 AM), https://www.law.com/international-edition/2023/10/09/macfarlanes-showed-us-how-its-using-harvey-ai-heres-what-we-learnt [http://perma.cc/GXH9-PSEL].

5. As perhaps most famously expressed by Joseph Schumpeter when he wrote: "The capitalist achievement does not typically consist in providing more silk stockings for queens but in bringing them within the reach of factory girls in return for steadily decreasing amounts of effort." JOSEPH A. SCHUMPETER, CAPITALISM, SOCIALISM, AND DEMOCRACY 67–68 (Routledge 2003) (1942).

6. DARON ACEMOGLU & SIMON JOHNSON, POWER AND PROGRESS 39–98 (2023) (describing, e.g., crop rotation as a technology that improved agricultural efficiency and yields but primarily benefited those who already owned land and had the resources to implement these new methods, the First Industrial revolution as enriching factory owners and entrepreneurs but at the expense of artisans and workers, and railroads and telecommunications developments as tending to concentrate wealth and power in the hands of who controlled those networks).

7. For discussion and cites, see, e.g., Colleen V. Chien, *The Inequalities of Innovation*, 72 EMORY L.J. 1, 13–16, 21 (2022), which provides examples of the large share of Americans for whom prescription drug medicines are not taken because they are not affordable, the inaccessibility of naloxone to treat opioid overdoses, the lack of research support for tropical diseases that impact poor populations, and the under-representation of poor and marginalized populations in innovation, impacting its direction and impact.

8. Xavier Jaravel, *The Unequal Gains from Product Innovations: Evidence from the U.S. Retail Sector*, 134 Q.J. ECON. 715, 717 (2019) (demonstrating how, empirically and theoretically, high-income households experienced a faster increase in product variety and lower inflation than did low-income households due to firms becoming increasingly attuned to the needs of high-end consumers in the face of rising inequality).

9. *See infra* Part I.

10. *See infra* Part I.

202 legal aid professionals in the fall of 2023 to gauge their use and attitudes toward AI tools. We next conducted a field experiment[11] in which ninety-one participants were given free access to paid AI tools: ChatGPT-4, a general purpose tool that interacts with users through a chat interface and provides answers to a broad range of requests and questions; Gavel, a legal document automation platform built on traditional AI and automation technologies; and Co-Counsel, billed as a "comprehensive legal assistant" for carrying out legal research and drafting tasks.[12] We also provided a subset of pilot participants, randomly chosen, with "concierge services," including office hours and weekly emails with use cases aimed at reducing the costs of becoming familiar with the tools.

At the end of the trial, we asked participants to describe their use patterns, experiences, and intents with respect to the continued use of tools. We also asked them to disclose in detail their beneficial use cases and to indicate, for each one, the estimated amount of time saved as well as their likelihood of reusing each use case in the future.

Our study yielded several findings about the introduction and use of generative AI tools by legal aid lawyers:

- Pilot participants generally reported productivity gains and the intent to continue use: 90% of pilot participants indicated some level of productivity increase in connection with the tools; 25% indicated a medium to high level of productivity increase; and 75% signaled their intent to continue using generative AI tools.[13]
- To address AI concerns, lawyers focused on lower-risk tasks: Pilot participants focused on lower-risk tasks like document summarization, preliminary or confirmatory legal research, translation from legalese or English into more accessible forms of communication, and getting to a first draft of a legal document, and reported a 50% productivity gain across such tasks.[14]
- A gender gap in organic uptake was not replicated in outcomes after using generative AI tools: Men in our baseline survey were two to three times (47% v. 17%) more likely to be using the

11. The exact duration of access varied by participant.
12. *See infra* Part I and Part II.
13. Conditional upon being given access. *See infra* Part III.
14. *See infra* Part III.

tools than women; post-pilot, the outcomes between men and women on a large range of outcomes were statistically indistinguishable.[15]

- Assistance improved outcomes: Pilot participants who received extra support services reported statistically significantly better outcomes than those that did not with respect to productivity, satisfaction, quality of output, frequency of use, attitudes, and intentions to continue use of paid tools.[16]

These results suggest that AI tools are poised to significantly aid legal professionals and narrow the justice gap, but also that how they are introduced matter—though women comprise the majority of public interest lawyers, organic uptake of generative AI was much higher among men in our study. Supportive services also appear to improve tool adoption. The participants' positive experiences support viewing AI technologies as augmenting rather than threatening the work of lawyers.[17] Legal aid lawyer-directed technological solutions may hold the greatest potential to dramatically increase the scope of coverage, and we suggest some steps, such as devising ways to certify the quality of legal information bots in the same way attorneys are bar-qualified, to support such generative collaborations. Along with the paper, we release a companion database of one hundred helpful use cases, including prompts and outputs, provided by legal aid professionals in the trial, to support broader adoption.

In Part I, we briefly describe the access to justice gap, past efforts to address it through technology, and relevant studies of the impact of generative AI on legal workers. Part II describes how we undertook our survey and pilot. Part III reports on the outcomes of our survey and pilot, and the resulting use case database we complied. Part IV discusses lawyers' concerns with AI tools and shows, through descriptions and examples of use cases, how they managed them to achieve productivity gains. Part V includes our observations and recommendations. To support broader dissemination and use of generative AI tools by legal aid lawyers, we release a compendium of approximately one hundred use cases produced as part of our pilot, available at https://bit.ly/AIA2J.

15. *See infra* Part III.
16. *See infra* Part III.
17. *See infra* Part III.

I. THE JUSTICE GAP AND ROLE OF AI

Though the widespread availability of generative AI tools is new, the justice gap, and efforts to close it through technology, are not.[18] There is also now an emerging literature considering the impact of generative AI on workers, including lawyers.[19] Our study of the role that generative AI can play in addressing the access to justice gap by enhancing legal aid workers sits at their intersection, which we briefly discuss to foreground our study.

A. The Justice Gap

The Legal Service Corporation estimates that 92% of substantial civil legal needs among low-income Americans are unmet.[20] This translates into millions of Americans lacking adequate legal help with problems in domains like housing, health care, and protection from abuse.[21] It is also estimated that over one hundred million Americans face civil justice problems pertaining to "basic human needs," including matters related to shelter, sustenance, safety, health, and child custody.[22]

Among the many causes of the access to justice gap, two are particularly salient. First, there is a knowledge gap, because many, if not most, individuals with civil legal needs do not know their problem is a legal issue or do not know how to get legal help.[23] Second, there is

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18. *See* Raymond H. Brescia et al., *Embracing Disruption: How Technological Change in the Delivery of Legal Services Can Improve Access to Justice*, 78 ALB. L. REV. 553, 588–611 (2015) (discussing the efforts to close the justice gap).

19. *See* discussion *infra* Section I.C.

20. LEGAL SERVS. CORP., THE JUSTICE GAP: THE UNMET CIVIL LEGAL NEEDS OF LOW-INCOME AMERICANS 8 (2022), https://lsc-live.app.box.com/s/xl2v2uraiotbbzrhuwtjlgi0emp3myz1 [https://perma.cc/YWR7-V8K7]. The Legal Services Corporation ("LSC") defines the "justice gap" as "the difference between the civil legal needs of low-income Americans and the resources available to meet those needs." *Id.* at 14.

21. *Id.* at 15. The LSC reported in 2022 that LSC-funded organizations are unable to provide any or adequate legal help for approximately 1.4 million (or 71%) civil legal problems that come to their doors every year. *Id.* at 75.

22. COMM'N ON FUTURE OF LEGAL SERVS., AM. BAR. ASS'N, REPORT ON THE FUTURE OF LEGAL SERVICES IN THE UNITED STATES 11–14 (2016), https://www.americanbar.org/content/dam/aba/images/abanews/2016FLSReport_FNL_WEB.pdf [https://perma.cc/S4HR-UQ9C] (quoting Rebecca L. Sandefur, *What We Know & Need to Know About the Legal Needs of the Public*, 67 S.C. L. REV. 433, 466 (2016)).

23. *See, e.g.*, ROCÍO AVALOS ET AL., STATE BAR OF CAL., 2019 CALIFORNIA JUSTICE GAP STUDY: EXECUTIVE REPORT 7 (2019), https://www.calbar.ca.gov/Portals/0/documents/accessJustice/Justice-Gap-Study-Executive-Summary.pdf [http://perma.cc/PMY7-Z7EE]. The access to justice gap is not the same thing as the access to lawyer gap, and some have suggested that skepticism about the value of legal services by low and middle-income consumers also explains why there is

a service gap.[24] Paid legal assistance is often out of reach and legal aid organizations can only serve about 30% of the clients that come through their doors.[25] While generative AI has the potential to address these and other problems, the focus of our study is on augmenting the capacity of legal aid organizations to meet the demands of clients. Three realms of legal assistance—eviction defense, expungement, and immigration—illustrate the scope and severity of the conditions that follow inadequate representation.

Each year, approximately 2.7 million households (or 7.6 million people) in the United States face the threat of displacement from their homes through court-ordered eviction.[26] Nearly 40% (or 2.9 million) of individuals facing the threat of eviction each year are children.[27] There is strong evidence that legal representation improves outcomes and housing stability for tenants in eviction or unlawful detainer proceedings.[28] For example, the Shriver Report (2020) found that representation by counsel in unlawful detainer proceedings in California led to close to zero defaults.[29] Even when landlords prevailed, 83% of represented tenants were able to secure negotiated settlements.[30] One year later, 71% of the represented tenants had secured a new rental unit, compared to 43% of tenants without an attorney.[31] But only 4% of tenants on average are represented by an attorney in eviction proceedings, as compared to 83% of landlords.[32]

Or, take the realm of expungement. An estimated one in three American adults has a criminal record, which can lead to barriers to

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a lack of access to justice. Milan Markovic, *Juking Access to Justice to Deregulate the Legal Market*, GEO. J. LEGAL ETHICS 63, 67 (2016).

24. Avalos et al., *supra* note 23, at 7.

25. *See* LEGAL SERVS. CORP., *supra* note 20, at 75.

26. Nick Graetz et al., *A Comprehensive Demographic Profile of the US Evicted Population*, PROC. NAT'L ACAD. SCIS., Oct. 2, 2023, at 1.

27. *Id.* at 3 (based on review of eviction cases from 2007 to 2016 and linking them to Census records).

28. JUD. COUNS. OF CAL., FINAL REPORT FROM THE WORK GROUP ON HOMELESSNESS TO THE CHIEF JUSTICE 8–9 (2021), https://www.courts.ca.gov/documents/hwg_work-group-report.pdf [https://perma.cc/S7M2-CD2X] (summarizing 2020 Shriver report and other studies); STOUT RISIUS ROSS, LLC, THE ESTIMATED COST OF AN EVICTION RIGHT TO COUNSEL OUTSIDE OF NEW YORK CITY 10–12 (2022) (describing benefits of representation during eviction proceedings).

29. Jud. Couns. of Cal., *supra* note 28, at 8.

30. *Id.*

31. *Id.* at 8–9.

32. NAT'L CTR. FOR ACCESS TO JUST., EVICTION REPRESENTATION STATISTICS FOR LANDLORDS AND TENANTS ABSENT SPECIAL INTERVENTION (2024), http://civilrighttocounsel.org/uploaded_files/280/Landlord_and_tenant_eviction_rep_stats__NCCRC_.pdf [https://perma.cc/Q4R5-FES9].

employment, housing, and related opportunities.[33] As a result, there is a significant earnings impact associated with living with a conviction.[34] But only a fraction of people eligible for expungement of their records have received the relief to which they are entitled under the law.[35] A 2020 study by one of us estimated, based on the examination of criminal records nationwide, that twenty to thirty million Americans eligible for expungement remedies had not received them.[36] Follow-on work by the Paper Prisons Initiative has documented uptake rates of less than 10% with respect to clearable convictions across a number of states.[37] The challenges of getting relevant records, determining eligibility, and following the relevant procedures are presently difficult to navigate without legal assistance, leaving millions of Americans in the "second chance gap."[38]

Immigration proceedings in the United States can result in particularly grave outcomes, including detention and removal and the separation of families.[39] Nevertheless, there is generally no right to counsel in immigration proceedings. While the government is always represented by counsel, approximately 40% of individuals are represented in immigration proceedings.[40] Evidence shows that immigrants

33. Colleen V. Chien, *America's Paper Prisons: The Second Chance Gap*, 119 MICH. L. REV. 519, 528 (2020); Colleen V. Chien & Miriam Kim, Use Case Database no. 102, https://bit.ly/AIA2J (last visited Mar. 22, 2024).

34. Colleen V. Chien et. al., *Estimating the Earnings Loss Associated with a Criminal Record & Suspended Driver's License*, 64 ARIZ. L. REV. 675, 675 (2022). The average lost earnings per year associated with a felony or misdemeanor conviction record is $6,400 and $5,100, respectively. *Id.*

35. *Id.*

36. Chien, *supra* note 33, at 523–24 (this estimate did not take into account criteria that are not readily ascertainable based on background check records, such as out of record fines and fees.).

37. *What Is the "Second Chance Gap"?*, PAPER PRISONS INITIATIVE, https://paperprisons.org/SecondChanceGap.html [https://perma.cc/X3GU-ALU6].

38. *See* Chien, *supra* note 33, at 543. Since 2020, 9 states have passed Clean Slate legislation to narrow the second chance gap, bringing to 12 the number of states that have enacted laws to automate expungements. *See* THE CLEAN SLATE INITIATIVE, https://www.cleanslateinitiative.org/states [https://perma.cc/H52J-N5R4].

39. CAL. COAL. FOR UNIVERSAL REPRESENTATION, CALIFORNIA'S DUE PROCESS CRISIS: ACCESS TO LEGAL COUNSEL FOR DETAINED IMMIGRANTS 1 (2016), https://www.nilc.org/wp-content/uploads/2016/06/access-to-counsel-Calif-coalition-report-2016-06.pdf [https://perma.cc/S3SZ-AAMP].

40. EXEC. OFF. FOR IMMIGR. REV., U.S. DEP'T OF JUST., ADJUDICATION STATISTICS: CURRENT REPRESENTATION RATES (Oct. 12, 2023), https://www.justice.gov/media/1174836/dl?inline [https://perma.cc/3ZZH-FAEB]. In 2018, Vera reported that almost half of all immigration court cases have gone unrepresented. KAREN BERBERICH & NINA SIULC, VERA INST. OF JUST., WHY DOES REPRESENTATION MATTER? THE IMPACT OF LEGAL REPRESENTATION IN IMMIGRATION COURT 1 (2018), https://www.vera.org/downloads/publications/why-does-representation-matter.pdf [http://perma.cc/VHG3-LRUK]. Based on a review of over 1.2 million deportation

represented by counsel are more likely to seek and obtain relief[41] and success in removal proceedings.[42] Never-detained respondents with counsel were about five times more likely to succeed; released respondents with counsel were three and a half times more likely to obtain relief, and detained respondents with representation were over two times more likely to succeed than unrepresented respondents.[43] Over the six-year period reviewed by the American Immigration Council, only 2% of those in removal proceedings who applied for relief were successful without counsel.[44] While correlation is not causation, and even the best lawyer cannot guarantee victory, in adversarial contexts, the absence of counsel is generally associated with worse outcomes.[45]

Though the details of the relevant proceedings, as well as the roles of lawyers,[46] vary, in most cases, legal aid lawyers are required to oversee multi-step processes that are repeated over multiple cases. This provides an opportunity for technological solutions that can scale across clients. Augmenting the capacity of lawyers so that they can serve a greater number of people and serve them more effectively is an important component of addressing the justice gap.

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cases decided between 2007 and 2012, the American Immigration Council reported in 2016 that only 37% of all immigrants, and only 14% of detained immigrants, secured legal representation in their removal cases. Ingrid Eagly & Steven Shafer, *Access to Counsel in Immigration Court*, AM. IMMIGR. COUNCIL 70 fig. 19 (Sept. 28, 2016), https://www.americanimmigrationcouncil.org/research/access-counsel-immigration-court [https://perma.cc/TH6W-J5N9] (only 37% of immigrants in removal proceedings are represented by counsel); *see also* Ingrid V. Eagly & Steven Shafer, *A National Study of Access to Counsel in Immigration Court*, 164 U. PA. L. REV. 1, 7 fig. 1 (2015) (63% of immigrants went to court without counsel).

41. *See, e.g.*, Eagly & Shafer, *Access to Counsel in Immigration Court*, *supra* note 40, at 70 fig. 19 (including release from detention, with 44% of represented respondents released after custody hearing, compared to 11%).

42. *Id.* at 50 fig. 14.

43. *Id.* at 51 fig. 15.

44. *Id.* at 76; *see also* CAL. COAL. FOR UNIVERSAL REPRESENTATION, *supra* note 39, at 3 (finding that "detained immigrants who had counsel succeeded *more than five times* as often as did their unrepresented counterparts").

45. *See, e.g.*, Mitchell Levy, *Empirical Patterns of Pro Se Litigation in Federal District Courts*, 85 U. CHI. L. REV. 1819, 1865 (2018) ("pro se plaintiffs are less than one-tenth as likely to win cases as represented plaintiffs, whereas pro se defendants are only about one-third as likely to win cases as represented defendants."). *But see* BENJAMIN H. BARTON & STEPHANOS BIBAS, REBOOTING JUSTICE: MORE TECHNOLOGY, FEWER LAWYERS, AND THE FUTURE OF LAW 107 (2017) (lawyers did not significantly outperform self-represented litigants in securing wins in non-complex civil cases.).

46. For example, routine "transactional" tasks like registering trademarks, filing business formation documents, developing simple prenuptial agreements, and filing for expungements in some case need little to no individualized attention from lawyers as compared to, for example, court appearances. Drew Simshaw, *Toward National Regulation of Legal Technology: A Path Forward for Access to Justice*, 92 FORDHAM L. REV. 1, 10–12, 16–17 (2023).

B. Historical Efforts to Address the Justice Gap Through Technology

Efforts to address the justice gap through technology have generally followed the more general development of tools to increase lawyer efficiency, particularly in the areas of legal research and e-discovery.[47] The Internet led to the creation of online platforms and services that provide legal information and assistance directly to the public, such as legal databases, websites, blogs, and self-help guides. Online legal services providers such as LegalZoom, Nolo, and RocketLawyer began emerging in the 2000s to provide consumers with legal information and more affordable, automated options for the preparation of some common legal documents such as wills, trademark registrations, and business formation documents.[48]

In 2000, the Legal Services Corporation also created a new Technology Initiative Grants (TIG) program, made possible through special funds appropriated by Congress, to encourage innovation in the delivery of legal assistance to low-income Americans.[49] As of this date, LSC has awarded over $86 million in TIG funding to LSC-funded providers.[50] One such program is A2J Author, an online program created in 2004 with the help of TIG funding.[51] A2J Author helps courts and legal aid organizations create "guided interviews" to help pro se litigants draft and file legal documents.[52]

The rise of AI and automation technologies over the last decade has spawned a new generation of innovative solutions designed to make legal services generally more accessible, with lawyers and consumers in mind. Lawyers have used AI and automation to carry out

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47. For example, one of the most notable technological advances for lawyers was the introduction of Shepard's on Lexis.com in 1999, replacing the cumbersome process of finding and validating legal authority using Shepard's in print. LEXISNEXIS, HOW TO SHEPARDIZE 3 (1999), https://lexisnexis.com/infopro/training/reference/lexisdotcom/shep.pdf [https://perma.cc/QCS6-7ZWR] (noting that "your legal-research instructor may prefer that you master SHEPARD'S in print before you learn SHEPARD'S on *lexis.com*").

48. *See, e.g.*, Raymond H. Brescia et al., *supra* note 18, at 569–75.

49. Sherley E. Cruz, *Coding for Cultural Competency: Expanding Access to Justice with Technology*, 86 TENN. L. REV. 347, 354 (2019).

50. Press Release, Legal Servs. Corp., Legal Services Corporation Awards $5.1 Million in Technology Grants to 29 Legal Aid Organizations (Nov. 16, 2023), https://www.lsc.gov/press-release/legal-services-corporation-awards-51-million-technology-grants-29-legal-aid-organizations [https://perma.cc/2R37-LRHC].

51. Brescia et al., *supra* note 18, at 599; *Chapter 1: A2J Author Overview*, A2J AUTHOR (May 2023), https://www.a2jauthor.org/content/chapter-1-a2j-author-overview [https://perma.cc/PL4Q-QBXY].

52. *Chapter 1: A2J Author Overview*, *supra* note 51.

tasks like streamlining legal research, electronic discovery, due diligence review, contract analysis, and other tasks.[53] For example, TextIQ, founded in 2014, leveraged machine learning and natural language processing to help lawyers reduce the time and cost of reviewing documents for responsiveness and privilege.[54] On the consumer-facing side, AI and automation technologies have been used to build chatbots and virtual kiosks that can walk self-represented litigants through a series of questions, help them identify their legal issue, connect them to a legal service provider, or help them draft a legal document.[55]

The advent of consumer-facing generative AI tools has ushered in new optimism about the potential for AI tools to transform the practice of law, including the access to justice gap. As Chief Justice John Roberts remarked as part of his 2023 Year-End Report on the Judiciary:

> For those who cannot afford a lawyer, AI can help. It drives new, highly accessible tools that provide answers to basic questions, including where to find templates and court forms, how to fill them out, and where to bring them for presentation to the judge—all without leaving home. These tools have the

53. *See, e.g.*, Frank Pasquale & Glyn Cashwell, *Four Futures of Legal Automation*, 63 UCLA L. REV. DISCOURSE 26, 34 (2015) ("[p]redictive coding" can "decrease time spent in discovery by 75%."); Agnieszka McPeak, *Disruptive Technology & the Ethical Lawyer*, 50 U. TOL. L. REV. 457, 466 (2019) (legal technology can "streamline legal-related tasks" and lead to "more accurate results, for less cost, and in a much quicker timeframe."). For example, ROSS Intelligence, founded by two AI researchers in 2014, used AI to offer end users the ability to ask questions using natural language rather than a keyword or Boolean search. Nina Haikara, *AI Success Story: U of T's ROSS Intelligence Returns to Toronto to Open New Research Headquarters*, UNIV. TORONTO (Mar. 29, 2017), https://www.utoronto.ca/news/ai-success-story-u-t-s-ross-intelligence-returns-toronto-open-new-research-headquarters [https://perma.cc/A3UG-7A2G]. The company shut down its business operations in 2020 but is continuing to defend against a contract and copyright lawsuit filed by Thomson Reuters. David Thomas, *ROSS Intelligence Hopes for Second Act After Blaming Thomas Reuters for Forced Shutdown*, WESTLAW TODAY (Dec. 11, 2020), https://today.westlaw.com/Document/I257808803bf211ebb3c1929a5b9ad316/View/FullText.html?transitionType=SearchItem&contextData=(sc.Default) [https://perma.cc/485G-CFJK].

54. *Relativity Acquires Text IQ to Drive Leadership in AI for e-Discovery, Compliance and Privacy*, PR NEWSWIRE (May 27, 2021), https://www.prnewswire.com/news-releases/relativity-acquires-text-iq-to-drive-leadership-in-ai-for-e-discovery-compliance-and-privacy-301300405.html [https://perma.cc/HH7Y-C7MK].

55. Cruz, *supra* note 49, at 364.

> welcome potential to smooth out any mismatch between available resources and urgent needs in our court system.[56]

While the deployment of generative AI technologies across legal domains is still in its infancy, several risks are already readily apparent. First, that technology will exacerbate existing inequities given that the capital to invest in technology requires, well, capital.[57] Drew Simshaw, for example, has warned of a two-tiered system in which those who can effectively harness legal AI or retain expensive human lawyers receive superior services, while others are left with inferior AI-driven assistance[58]—a dehumanized version of the law.[59] The *Mata v. Avianca* suit, in which a lawyer infamously cited sources in court pleadings that were hallucinated by ChatGPT, provides one example.[60] Though receiving less attention, the posture of the parties is note-worthy: the case involved a personal injury claim brought by a plaintiff represented by a small practice[61] that improperly relied on ChatGPT to do his work, pitted against a large corporation, Avianca S.A.[62] As low-resourced plaintiffs turn to lawyers who promise cost savings based on their use of generative AI tools, or use them themselves, a two-tier model may emerge. Limited internet access and algorithmic literacy among low-income individuals could further hinder their access to innovative legal services.[63]

The risk of unfair and discriminatory outcomes also looms. Racial and gender biases have been widely documented in contexts that range

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56. HON. JOHN G. ROBERTS, JR., U.S. SUP. CT., 2023 YEAR-END REPORT ON THE FEDERAL JUDICIARY 5–6 (2023), https://www.supremecourt.gov/publicinfo/year-end/2023year-endreport.pdf [https://perma.cc/MQ4N-H9EV] ("As AI evolves, courts will need to consider its proper uses in litigation. In the federal courts, several Judicial Conference Committees . . . will be involved in that effort.").

57. See examples provided in ACEMOGLU & JOHNSON, *supra* note 6, at 40.

58. *See* Drew Simshaw, *supra* note 46, at 12–13.

59. HON. JOHN G. ROBERTS, JR., *supra* note 56 at 5 (describing AI as risking "dehumanizing the law").

60. Mata v. Avianca, Inc., No. 22-CV-1461, 2023 WL 4114965, at *9 (S.D.N.Y. June 22, 2023).

61. The lawyer's firm, Levidow, Levidow and Oberman, does not appear to have a website and is indicated on their website LinkedIn page to have 2–10 employees. *Id.* at *1; Levidow, Levidow & Oberman, P.C., LINKEDIN, https://www.linkedin.com/company/levidow-levidow-&-oberman-p.c./about/ [perma.cc/NKA2-L8KL].

62. Billed as the second-largest airline in South America. *Avianca (AV/AVA) Airline Information*, FLIGHTERA.NET, https://www.flightera.net/en/airline/Avianca [https://perma.cc/7ZGN-WHCN].

63. Simshaw, *supra* note 46, at 15–16.

from facial recognition[64] to recidivism prediction,[65] and age bias claims have been alleged and settled in the context of hiring algorithms.[66] In 2023, five federal agencies, including the U.S. Equal Employment Opportunity Commission, the Federal Trade Commission, the Department of Justice, and the Consumer Financial Protection Bureau, launched a coordinated initiative to guard against the use of "inaccurate, biased, and discriminatory" AI tools and their application in realms like hiring, tenant screening, and credit decisions.[67] Bias can show up in unexpected ways as well. In one study, GPT-3 correctly answered 77% of the 200 questions in the study related to online terms of service.[68] GPT-3 was poor in answering pro-consumer questions, potentially reflecting an anti-consumer bias.[69] Performance was also highly sensitive to the wording of questions.[70] Based on these findings, Kolt concluded that, "while language models have the potential to empower consumers, they also have the potential to provide misleading advice and entrench harmful biases."[71] Concerns about data privacy, confidentiality, hallucinations, and fabrications are also urgent among the lawyers we polled and described further in Part IV.

Another set of problems relates to the risk of potential consumer harm through offerings that result in the unauthorized practice of law.[72] Unauthorized practice of law (UPL) rules generally prohibit

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64. Joy Buolamwini & Timnit Gebru, *Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*, PROC. MACH. LEARNING RSCH., Feb. 2018, at 1–12 (documenting the worse performance of facial recognition systems among women and people of color as compared to white males).

65. Jeff Larson et al., *How We Analyzed the COMPAS Recidivism Algorithm*, PROPUBLICA (May 23, 2016), https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm [https://perma.cc/6RYG-HVPT] (finding that black defendants were twice as likely as white defendants to be misclassified as a higher risk of violent recidivism, while white recidivists were more likely to misclassified as low risk).

66. Press Release, U.S. Equal Emp. Opportunity Comm'n, iTutorGroup to Pay $365,000 to Settle EEOC Discriminatory Hiring Suit (Sept. 11, 2023), https://www.eeoc.gov/newsroom/itutorgroup-pay-365000-settle-eeoc-discriminatory-hiring-suit [https://perma.cc/U485-JQFM].

67. Rohit Chopra et al., *Joint Statement on Enforcement Efforts Against Discrimination and Bias in Automated Systems*, U.S. EQUAL EMP. OPPORTUNITY COMM'N, https://www.eeoc.gov/joint-statement-enforcement-efforts-against-discrimination-and-bias-automated-systems [https://perma.cc/T2KF-SBKP].

68. Noam Kolt, *Predicting Consumer Contracts*, 37 BERKELEY TECH. L.J. 71, 136 (2022).

69. *Id.* at 106–08.

70. *Id.* at 80, 104, 107–09.

71. *Id.* at 71.

72. *See, e.g.*, Katherine L.W. Norton, *The Middle Ground: A Meaningful Balance Between the Benefits and Limitations of Artificial Intelligence to Assist with the Justice Gap*, 75 U. MIAMI L. REV. 190, 248 (2020); Spencer Williams, *Predictive Contracting*, 2019 COLUM. BUS. L. REV.

non-lawyers from engaging in the provision of legal advice unless they are authorized to do so in a particular jurisdiction.[73] This includes not only practicing law in violation of the regulations of the legal profession in that jurisdiction, but also assisting another in doing so.[74] While varying from jurisdiction to jurisdiction, these rules are designed to protect the public against the rendition of legal services by unqualified persons, a class that some may argue includes legal technologists who are not lawyers.[75]

Yet UPL rules have also been subject to criticisms that they are more about protecting the bar and lawyers than about protecting consumers.[76] There is a parallel and just as urgent risk, as Orly Lobel discusses in her book, *The Equality Machine*, that an undue focus on the harms of AI will result in a failure of AI to deliver on its potential benefits, particularly with respect to access to justice.[77] As Justice Roberts has quipped, "[t]he legal profession is, in general, notoriously averse to change."[78] This conservatism, when applied to technology, has meant that efforts to address access to justice challenges have historically "been hindered by resistance to technological changes and other innovations."[79] Several barriers can hinder the effective "calibration" of technological solutions to access to justice problems, including resource constraints, lack of resilience, and relationship challenges.[80] However, at the heart of these lie regulations that discourage or prohibit legal services providers from accessing capital for technology, which in many cases stem from fears of engaging in the unauthorized practice of law (UPL).[81]

Drew Simshaw has identified resource, resilience, and relationship barriers to the effective "calibration" of technological solutions to access to justice problems but has also commented that underlying

621, 630, 691–93 (2019) (discussing the potential unauthorized practice of law by AI contracting tools).

73. MODEL RULES OF PRO. CONDUCT r. 5.5 (AM. BAR ASS'N 2019).

74. *Id.*

75. MODEL RULES OF PRO. CONDUCT r. 5.5 cmt. 2 (AM. BAR ASS'N 2019).

76. NAT'L CTR. FOR ACCESS TO JUST., "UNAUTHORIZED PRACTICE OF LAW" ENFORCEMENT IN CALIFORNIA: PROTECTION OR PROTECTIONISM? 6, 12 (2022), https://ncaj.org/sites/default/files/2022-02/Cease%20and%20Desist%20Report%20-%20%20Final%2C%202-14-22%20pdf.pdf [https://perma.cc/CVQ3-7SLZ].

77. ORLY LOBEL, THE EQUALITY MACHINE 5–12 (2022).

78. HON. JOHN G. ROBERTS, JR., *supra* note 56, at 3.

79. COMM'N ON FUTURE OF LEGAL SERVS., *supra* note 22, at 8–9.

80. *See* Simshaw, *supra* note 46, at 15–20.

81. *Id.* at 19–20.

all of these barriers are regulations that discourage or prohibit legal services providers from accessing capital for technology, in many cases out of a fear of engaging in UPL.[82]

Utah, Arizona, and several other states have moved forward with regulatory reforms that mitigate UPL issues. Utah's regulatory sandbox, the first of its kind, allows participants to offer "innovative legal services" to the public "in a controlled environment" while enabling data collection and ongoing supervision to protect consumers from harm.[83] Arizona's regulatory reform efforts, without a sandbox per se, consist of the state adopting a framework that allows alternative business structures for law firms.[84] A recent study of these initiatives in Utah concluded that reform efforts were "spurring substantial innovation" in legal services delivery models without a "substantial risk of consumer harm."[85] A small number of states, including Utah, Arizona, Delaware, and Alaska, have also approved initiatives to allow nonlawyer community-based advocates to provide limited-scope legal advice in certain areas, such as family law, medical debt, and housing.[86] But other states have been slower to embrace reforms: while California,

82. *Id.* at 18–20.

83. HON. MATTHEW B. DURRANT, UTAH SUP. CT., UTAH SUPREME COURT STANDING ORDER NO. 15 (AMENDED) 2–3, 11 (Sept. 21, 2022). As of December 2023, the office overseeing the sandbox had received only 9 harm-related complaints, or approximately 1 complaint per 7,937 services delivered. UTAH OFF. OF LEGAL SERVS. INNOVATION, ACTIVITY REPORT: DECEMBER 2023 6 (2024), https://utahinnovationoffice.org/wp-content/uploads/2024/02/December-Activity-Report.pdf [https://perma.cc/NY8M-TBFC].

84. *Alternative Business Structures*, AZCOURTS.GOV, https://www.azcourts.gov/cld/Alternative-Business-Structure [https://perma.cc/2NTX-3UJ7] (abrogating Ethical Rule 5.4 and adopting new Rule 31.3(e)(4) effective January 1, 2021); *see also* TASK FORCE ON DELIVERY OF LEGAL SERVS., ARIZ. SUP. CT., REPORT AND RECOMMENDATIONS 10–11 (2019), https://www.azcourts.gov/Portals/74/LSTF/Report/LSTFReportRecommendationsRED10042019.pdf?ver=2019-10-07-084849-750 [https://perma.cc/J5DY-4W7B].

85. DAVID F. ENGSTROM ET AL., STANFORD L. SCH. DEBORAH L. RHODE CTR. ON LEGAL PRO., LEGAL INNOVATION AFTER REFORM: EVIDENCE FROM REGULATORY CHANGE 7, 37 (2022).

86. *See Housing Stability Legal Advocates*, INNOVATION FOR JUST., https://www.innovation4justice.org/hsla [https://perma.cc/Z89R-JSLG]; Press Release, Del. Sup. Ct., Delaware Supreme Court Announces Adoption of New Supreme Court Rule 57.1 to Allow Non-Lawyer Representation of Residential Tenants in Eviction Actions (Jan. 28, 2022), https://courts.delaware.gov/forms/download.aspx?id=133348 [https://perma.cc/J7TN-VZPN]; *Community Justice Worker Program*, ALASKA LEGAL SERVS. CORP., https://www.alsc-law.org/community-justice-worker-program/ [https://perma.cc/M8YG-ET49]; Tracey Read, *Ariz., Utah OK Nonlawyer Program for Housing Advice*, LAW360 (May 30, 2023, 4:14 PM), https://www.law360.com/access-to-justice/articles/1682740/ariz-utah-ok-nonlawyer-program-for-housing-advice [https://perma.cc/42V7-C5RD].

Washington, and Florida, have also explored regulatory sandboxes in the legal sector, none have been approved.[87]

For these reasons, it is likely that for the foreseeable future, legal technology will be intermediated through legal professionals, making it an opportune time to test the extent to which their work can be enhanced by generative AI technology. As we elaborate in the following sections, the current body of evidence, which serves as the foundation for our work, presents a complex picture.

C. Generative AI and the Legal Profession

A number of studies have considered the extent to which generative AI tools can be useful to legal professionals. The studies that examine the use of generative AI in the legal profession using OpenAI's GPT-3.5 emphasize the opportunities and limitations. Shortly after the launch of GPT-3.5, Perlman (2022) used it to generate an article that predicted generative AI's potential to revolutionize the legal industry by conducting legal research, generating legal documents, providing legal information, and performing legal analysis.[88] However, Perlman also found ChatGPT's responses were "imperfect and at times problematic" and warned that "the use of an AI tool for law-related services raises a host of regulatory and ethical issues."[89] These concerns are echoed by Dahl et al. (2024) which concluded, based on a study, that "legal hallucinations are alarmingly prevalent" in large language

87. California took steps towards forming a regulatory sandbox, but those efforts came to a halt in 2022. In 2020, the State Bar formed a Working Group on Closing the Justice Gap to explore the creation of a regulatory sandbox. But the working group was dissolved in 2022 in light of significant restrictions imposed by the annual fee bill (Assembly Bill 2958). *Archived: Closing the Justice Gap Working Group*, STATE BAR CAL., https://www.calbar.ca.gov/About-Us/Who-We-Are/Archived-Committees/Closing-the-Justice-Gap-Working-Group [https://perma.cc/QPP6-EYQV]. In Washington, the Practice of Law Board has recommended that the Supreme Court create a legal regulatory lab to foster innovation. WASH. CTS. PRAC. L. BD., BLUEPRINT FOR A LEGAL REGULATORY LAB IN WASHINGTON STATE 8–9 (2022), https://www.wsba.org/docs/default-source/legal-community/committees/practice-of-law-board/polb_legal-regulatory-lab_2.0_02-2022.pdf?sfvrsn=b67110f1_5 [https://perma.cc/3UFK-EJX7]. In 2021, a special committee of the Florida Supreme Court recommended establishing an innovation lab program similar to the Utah sandbox, but the proposal was rejected. *See* Gary Blankenship, *Board Rejects Special Committee's 'Legal Lab' Recommendation*, FLA. BAR (Dec. 7, 2021), https://www.floridabar.org/the-florida-bar-news/board-rejects-special-committees-legal-lab-recommendation/ [https://perma.cc/REL7-PH6G].

88. Andrew M. Perlman, *The Implications of ChatGPT for Legal Services and Society*, 30 MICH. TECH. L. REV. (forthcoming 2023) (manuscript at 2–3), https://ssrn.com/abstract=4294197 [https://perma.cc/DG5N-XW5N].

89. *Id.* at 1.

models (LLM), occurring 69% (ChatGPT 3.5) to 88% (Meta's Llama 2) of the time when asked specific questions about federal court cases.[90]

Another set of studies has been more sanguine about the prospect of generative AI's usefulness to the legal profession. For example, while one study showed that OpenAI's GPT-3.5 failed the bar exam, Katz et al. (2023) found that ChatGPT-4 obtained a score in the 90th percentile on the Uniform Bar Examination and a passing score in all jurisdictions.[91] Similarly, studies involving law school exams have shown greater benefits with more recent versions of generative AI tools. Choi et al. (2023) found that ChatGPT 3.5 averaged a C+ on four real law school examinations.[92] In general, ChatGPT performed better on essay questions than multiple choice questions, with some answers being above average and other answers being dramatically incorrect.[93] When Choi and Schwarcz tested GPT-4's impact on law students' final examination performance,[94] they found that GPT-4 significantly improved students' performance on multiple-choice questions, although it had no effect on essay answers. Similar to more general workplace studies,[95] the study showed evidence of productivity compression: the lowest-performing students performed significantly

90. MATTHEW DAHL ET AL., LARGE LEGAL FICTIONS: PROFILING LEGAL HALLUCINATIONS IN LARGE LANGUAGE MODELS 1 fig. 1 (2024), http://arxiv.org/abs/2401.01301 [https://perma.cc/JV88-497A].

91. Daniel Martin Katz et al., *GPT-4 Passes the Bar Exam*, PHIL. TRANSACTIONS R. SOC'Y A, Apr. 2024, at 1, 12 n.5. *But see* Eric Martínez, *Re-Evaluating GPT-4's Bar Exam Performance* 3 (Legal Priorities Project, Working Paper No. 2-2023, 2023), https://www.legalpriorities.org/documents/2023-2%20-%20Re-Evaluating%20GPT-4s%20Bar%20Exam%20Performance.pdf [https://perma.cc/4BKT-H3UP] (investigating methodological challenges in verifying claim that GPT-4 scored 90th-percentile on Uniform Bar Exam).

92. Jonathan H. Choi et al., *ChatGPT Goes to Law School*, 71 J. LEGAL EDUC. 387, 391 (2022) (discussing ChatGPT's performance on over 95 multiple choice questions and 12 essay questions in Constitutional Law: Federalism and Separation of Powers, Employee Benefits, Taxation, and Torts).

93. *Id.* at 391.

94. Jonathan H. Choi et al., *AI Assistance in Legal Analysis: An Empirical Study*, 73 J. LEGAL EDUC. (forthcoming, 2024) (manuscript at 1) https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4539836 [https://perma.cc/8MYE-ZSKH].

95. *See, e.g.*, Erik Brynjolfsson et al., *Generative AI at Work* 1, 15 (2023), https://www.nber.org/system/files/working_papers/w31161/w31161.pdf [https://perma.cc/D2XU-LY8A] (summarizing some of this literature and providing its own finding that agents with access to conversational guidance from a generative AI tool successfully resolved 14% more customer chats per hour, with less-experienced and lower-skill workers experiencing the greatest gains (34%) with a minimal impact on experienced and highly-skilled workers).

better when using GPT-4, while the best-performing students performed worse.[96]

Outside the exam context, several studies have examined the use of generative AI on hypothetical legal tasks. One study (Choi, Monahan & Schwarcz, 2023) conducted a randomized controlled trial of the use of AI by fifty-nine law students.[97] The students completed four hypothetical legal tasks (drafting a complaint, a contract, a section of an employee handbook, and a client memorandum), with or without GPT-4's assistance.[98] The study showed that the addition of GPT-4 consistently and uniformly reduced the amount of time students took to complete tasks, although it only slightly and inconsistently improved the quality of students' legal analysis.[99]

Some studies have used hypothetical legal tasks to evaluate the extent to which generative AI can replace lawyers. Trozze et al. (2023) prompted GPT-3.5 to identify potential legal violations when presented with fact patterns from real-life securities cryptocurrency cases.[100] Mock jurors were also asked to assess complaints written by either ChatGPT or a human lawyer.[101] The study showed that GPT-3.5 performed very well at drafting a complaint, but its legal reasoning proved weak as it missed certain violations.[102] Another study provided twenty-five legal professionals with a summary of a real-world employment case and asked them to review two versions of a letter from the employee's lawyer.[103] When asked which letter was more effective in presenting the case, 80% of participants preferred the letter written by ChatGPT-4 over the letter written by a trained lawyer.[104]

In sum, to date, studies of AI in the legal profession have focused on law school exams and/or fictional scenarios, rather than on actual

96. *Id.* at 15–16.

97. Jonathan H. Choi et. al, *Lawyering in the Age of Artificial Intelligence*, MINN. L. REV. (forthcoming 2024) (manuscript at 5).

98. *Id.*

99. *Id.* at 21.

100. ARIANNA TROZZE ET AL., LARGE LANGUAGE MODELS IN CRYPTOCURRENCY SECURITIES CASES: CAN A GPT MODEL MEANINGFULLY ASSIST LAWYERS? 1 (2023), https://arxiv.org/pdf/2308.06032.pdf [https://perma.cc/J45V-MMBB].

101. *Id.*

102. *Id.* at 26–28.

103. Lena Wrzesniowska, *Can AI Make a Case? AI Vs. Lawyer in the Dutch Legal Context*, INT'L J. L., ETHICS, & TECH. (forthcoming 2023) (manuscript at 10), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4614381 [https://perma.cc/YB2U-LTVU].

104. *Id.* at 26 (evaluating effectiveness of AI-generated and human lawyer-written letter based on persuasiveness, clarity, key arguments, and use of evidence).

use cases. To our knowledge, there are no studies involving the use of generative AI by practicing lawyers, much less legal aid lawyers. In addition, we are not aware of any studies—legal or otherwise—that have focused particularly on the impact and use of generative AI in public interest contexts. These contexts may give rise to particular needs, for example, for multilingual communications appropriate for different education levels and tasks like volunteer coordination, fundraising, and managing turnover in clients by design.[105]

We sought to fill this gap by conducting a field experiment of legal aid lawyers using AI tools to perform actual use cases in connection with their work. We modified the rollout for some participants by offering a subset, chosen randomly, with supportive services, based on literature that suggests that providing assistance in legal administrative settings can close success gaps.[106] By coupling our field experiment with a survey, we can also get a perspective on the "organic" uptake of the tools and its variance across dimensions.

II. METHODOLOGY

Our research was motivated by several questions, both descriptive and causal. First, to what extent are generative AI tools being used by the legal aid community and how are they perceived? Second, what happens when paid generative tools are provided to legal aid attorneys, in terms of their use, productivity, opinions, and intents with respect to generative AI tools? And third, does the way that tools are introduced (i.e., with or without support) impact the outcomes associated with sustained exposure to them through the pilot? Across these questions, we were also curious about the extent to which demographic and experience traits were correlated with survey respondents or pilot participants' experiences.

In this section, we describe the methodology we used to conduct the survey and field study we used to answer these questions, with human subjects' approval from the Committee for Protection of Human Subjects (CPHS) of the Office for Protection of Human Subjects (OPHS) at UC Berkeley.[107]

105. For example, the individuals that responded to our survey worked in an average of 4.5 legal areas (median = 3). *See infra* Part II.

106. *See* Nick A. Pairolero et al., *Closing the Gender Gap in Patenting: Evidence from a Randomized Control Trial at the USPTO* 3, 22 (U.S. Pat. & Trade Off. Econ. Working Paper Series, Paper No. 2022-1, 2022), https://ssrn.com/abstract=4265093 [https://perma.cc/EJ6T-Z983].

107. University of California Berkeley IRB Protocol 2023-10-16780.

A. Recruitment and Data Collection

We recruited participants from three sources. First, the State Bar of California Office of Access & Inclusion emailed 149 legal aid practitioners inviting them to take our baseline survey in connection with a convening hosted at the State Bar's Los Angeles office on October 26, 2023 (see Appendix Survey 1). The request to fill out the survey was reinforced during a presentation in person at the convening on AI use cases for legal aid lawyers. The survey included an invitation to participate in the pilot of AI/automation tools; those who expressed interest in the pilot received a link to a consent form. 118 individuals, or close to 80% of those who were sent the survey completed it, and we enrolled 51 participants in the pilot through this initial outreach.[108]

Second, in November 2023, we distributed a second round of the same survey by email to our informal networks of legal aid professionals and also invited survey recipients and staff at the California State Bar Office of Access & Inclusion and the Legal Services Corporation to share the survey with their networks. Third, we emailed the baseline survey to individuals recommended by their peers from the first distribution. Eighty-six people completed the survey, and 40 participants joined the pilot based on one of these two forms of outreach.

Our baseline survey (available in the Appendix), which each person in both the survey group and pilot group took, asked about the relevant experiences of each respondent.[109] Those who accepted our invitation to join the pilot and consented to research were welcomed into the pilot from November 3, 2023, to November 29, 2023.[110]

At the conclusion of the pilot, we asked pilot participants to fill out a second exit survey that asked similar questions to the ones that appeared in the baseline, but we also asked for additional details regarding each person's experiences in the pilot, changes as a result of the pilot, and intentions for future use of AI tools.

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108. Twenty-six survey takers, and the remainder from people at the organizations of survey takers who were referred by them.

109. As detailed in the appendix, there were slight differences between the first and second distribution of the baseline survey. When asked about the number of concerns, use cases, and potential use cases of generative AI in the second distribution of the baseline survey, the limitation "choose your top 3" was introduced. Taking this into account, whenever we made direct comparisons between the baseline and exit survey, we restricted our baseline survey results to the first distribution, where the wording was the same. *See infra* Appendix.

110. Although, dates of ChatGPT-4 access varied due to company developments around that time.

Two hundred and two people completed the initial survey, and 91 people began the pilot. Over the course of the pilot, 10 participants dropped out, citing time constraints or failing to respond to our email asking for verification of a ChatGPT account that we could upgrade. Eighty-one finished the pilot, and 66 completed the exit survey, a response rate of 81%.

Of the 91 people who began the pilot, 54 were placed in the control group and 37 in the group treated with concierge services. At the conclusion of the pilot, we were left with 51 people in the control group and 30 in the treated group. As part of exit processing, participants were asked to fill out a final survey and, separately, provide use cases for future sharing with their permission. About 80% of each of the concierge and control group[111] responded to the survey, and about 41 pilot participants provided use cases for further sharing.

B. Pilot Design

We structured our pilot as a field trial in which participants were offered generative AI services, and then a subset of participants, chosen randomly, were provided with additional AI support services. Our pilot included three AI tools that include automation, "traditional AI," and generative AI capabilities and that could, in principle, be used by lawyers or consumers.

Gavel is a platform for automating legal documents and workflows. The service has been described as a no-code technology that enables "subject-matter experts to build data collection and intake applications that generate . . . rules-based legal workflows and documents."[112] Founded in 2018, the platform enables both law firms as well as nonprofits and pro bono lawyers to build client-facing workflows and document templates for business clients, domestic violence survivors, low-income tenants, and other clients.[113] Its website describes its ability to help streamline attorney-client interactions,

111. 41 from the control group and 25 from the concierge group.

112. *Documate Rebrands as Gavel to Expand Its Platform for Legal Products; Adds Pierre Martin as CTO*, GLOBENEWSWIRE (Jan. 31, 2023, 11:00AM), https://www.globenewswire.com/en/news-release/2023/01/31/2598772/0/en/Documate-Rebrands-as-Gavel-to-Expand-Its-Platform-for-Legal-Products-Adds-Pierre-Martin-as-CTO.html [http://perma.cc/3J5T-CHB8].

113. *Gavel Co-Founder, Dorna Moini—In-depth Interview*, ARTIFICIAL LAW. (Feb. 19, 2021), https://www.artificiallawyer.com/2021/02/19/documate-co-founder-dorna-moini-in-depth-interview/ [http://perma.cc/7L47-TVV7]; Gavel, LINKEDIN, https://www.linkedin.com/company/gaveltech/about/ [perma.cc/U8R9-FDKE].

customize tools, and manage intake across a variety of legal domains, including estate planning, family law, bankruptcy, and IP, and it also describes the use of AI that helps carry out automation.[114] The standard package was priced, around the time of the trial, at $165/month per seat. It also offered non-profit and educational rates.[115]

ChatGPT-4 is the premium version of the conversational chatbot offered by OpenAI. It is a general-purpose generative AI tool that responds to natural language prompts and is able to answer questions, write copy, draft emails, provide code, explain concepts, and provide assistance in many other contexts. As one source put it, "[i]t's a chatbot, but a really, really good one."[116] During the course of the trial, OpenAI introduced the capability to create custom GPTs that are capable of being pre-programmed with customized information. At the time of the trial, ChatGPT-4 was priced at twenty dollars per month per person.

CoCounsel is a generative AI legal platform powered by OpenAI's large language model GPT-4 and Casetext's proprietary legal databases. Customized for legal applications, CoCounsel is advertised as capable of being able to "read, comprehend, and write at a postgraduate level" and to perform a variety of legal tasks, including legal research, document review, deposition preparation, and contract analysis.[117] Casetext's website states that CoCounsel is more secure than ChatGPT, insofar as user data provided to CoCounsel is not used to train the AI model.[118] At the time of our trial, CoCounsel was being offered at a retail price of approximately $400 per seat per month.[119]

Baseline Treatment

All participants were offered free, one-month subscriptions to OpenAI's ChatGPT-4[120] and Gavel.[121] The participants who worked

114. *Plans & Pricing*, GAVEL https://www.gavel.io/pricing [https://perma.cc/826E-P6NE].

115. *Id.*

116. Harry Guinness, *How Does ChatGPT Work?*, ZAPIER (Sept. 6, 2023), https://zapier.com/blog/how-does-chatgpt-work/ [https://perma.cc/6DXJ-AALE].

117. *CoCounsel,* CASETEXT, https://casetext.com/cocounsel/ [https://perma.cc/XJ9D-Z2ZK].

118. *Security*, CASETEXT, https://casetext.com/security/ [https://perma.cc/QP4T-4YJK].

119. *Subscription Plans,* CASETEXT, https://web.archive.org/web/20231126213602/https://casetext.com/subscribe [perma.cc/S67B-PP8F].

120. *ChatGPT Login*, OPENAI, https://chat.openai.com/auth/login [https://perma.cc/58F7-YQWF]; *see infra* Appendix, Survey 1.

121. GAVEL, https://www.gavel.io/ [https://perma.cc/GJ6L-JCCW]; *see infra* Appendix, Survey 1.

at legal aid organizations funded by the State Bar of California Office of Access & Inclusion[122] also received free subscriptions to Casetext's CoCounsel for approximately four weeks, although some had their access extended further.[123] We also invited users to try the free versions of Anthropic's Claude,[124] Google Bard (recently renamed Gemini),[125] and Microsoft Bing Chat,[126] all large-language models that provide conversational interfaces similar to ChatGPT.[127]

RCT

We implemented our study of the impact of concierge services as a randomized controlled trial (RCT), randomizing at the individual enrollee level. To minimize spillovers between groups, we placed all participants from a single organization into the same (control or treatment) group. The "concierge" support was in the form of

1. weekly emails with sample use cases and links to tailored training;[128]
2. invitations to weekly office hours hosted three days a week by one of the authors and a trained third-year law student;[129]
3. invitations to tailored trainings presented by representatives of Gavel and Casetext; and
4. access to curated training materials, including slides and/or recordings of the tailored training sessions.

We provided members of the control group subscriptions to the tools but did not offer them any additional concierge support; they were largely left to their own devices. However, some members of the control group attended AI presentations at a State Bar event as discussed above.

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122. *See Grants*, STATE BAR CAL., https://www.calbar.ca.gov/Access-to-Justice/Grants [https://perma.cc/3WSF-RJPK]; *2024 Legal Aid Grant Recipients*, STATE BAR CAL. (2024), https://www.calbar.ca.gov/Portals/0/documents/accessJustice/Legal-Aid-Grant-Recipients.pdf [http://perma.cc/4JFF-UPUR] (2024 list of grantees).

123. Tools described in more detail in Part I.

124. CLAUDE, https://claude.ai/login?returnTo=%2F [https://perma.cc/P6M7-BJHG].

125. GEMINI, https://gemini.google.com/ [https://perma.cc/96DC-A9QK].

126. *Copilot*, MICROSOFT BING, https://www.bing.com/chat [https://perma.cc/7VL7-9YTR].

127. *See supra* notes 124–26.

128. Use cases that were shared included summarizing documents, writing grant applications, automating letters, and building a custom GPT.

129. Office hours were held by Zoom using a Berkeley Zoom account on Mondays at 12 p.m. (PST) and Wednesdays and Fridays at 9 a.m. (PST).

C. Analysis Approach and Respondent Profile

Once we had our initial results, missing data for demographic variables (age, race, gender, experience level) on the initial survey were imputed with available values from the pilot survey, and vice versa. Where demographic data was missing, we used LinkedIn disclosures to supplement the information we had regarding relevant work experience (which we inferred based on year of graduation from law school) in combination with visual inspection and probabilistic classifiers to infer binary gender and race. We worked with a statistician to perform a series of descriptive and regression analyses on the resulting data, the details of which are provided in the Appendix, along with information about the demographic profile of our respondents and participants.[130] In the next Part, we report the results of our analyses, noting differences that were statistically significant.

The "average" respondent to our survey was a white, female lawyer with roughly 10–15 years of experience in the field, in her late 30s to early 40s.[131] The average number of practice areas was 4.5 (median = 3). Our responses skewed female, consistent with the demographics of legal aid practitioners:[132] among people for whom we had binary gender data, 78% of the respondents to our baseline survey and 69% of the participants to our pilot study were women.[133] As compared to the average respondent to our survey, the average participant in our pilot was younger, more likely to be male, and less experienced.[134] The greatest share of lawyers in this population had

130. *STATA Release 15*, STATA, https://www.stata.com/stata15/ [https://perma.cc/TS5N-7QKH].

131. *See infra* Appendix, Demographics of Survey Takers.

132. 73.8% of all full-time staff and 77.4% of part-time staff at Legal Services Corporation grantees were women in 2022. LEGAL SERVS. CORP., 2022 BY THE NUMBERS: THE DATA UNDERLYING LEGAL AID PROBLEMS 72 (2023), https://lsc-live.app.box.com/s/h2bajpr3gps4s4a1iio6fwiddhmu1nwb [https://perma.cc/B9MU-E8AF]. In comparison, men outnumber women in the law in California. *Diversity of 2022 California Licensed Attorneys*, STATE BAR CAL. (2022), https://publications.calbar.ca.gov/2022-diversity-report-card/diversity-2022-california-licensed-attorneys [https://perma.cc/Z5RX-SDQ4] (44% of lawyers in California identified as female in 2022). *See also* Catherine Albiston et al., *Making Public Interest Lawyers in a Time of Crisis: An Evidence-Based Approach*, 34 GEO. J. LEGAL ETHICS 223, 264 (2021) (finding that "identifying as female is predictive of public interest work at later stages of one's career but not at the beginning . . . gender *is* significantly related to holding a *current* job in a public interest setting"); *id.* at 239 n.80 (summarizing past research finding that women law students prefer to pursue public interest work).

133. *See infra* Appendix, Demographics of Survey Takers.

134. *See infra* Appendix, Demographics of Survey Takers.

0–4 years of experience,[135] consistent with the finding described previously in Part I that tools can be most helpful for the least experienced.

III. SURVEY AND PILOT RESULTS

A. Patterns of Use and Perceptions of the Benefits and Risks of Generative AI Tools

We first sought to determine, as a descriptive matter, the extent of use of generative AI tools by legal aid professionals. Around 21% of participants overall reported that they were currently using the tools in their work. This figure was in line with the result of a comparable survey reported several months earlier by LexisNexis that found that 16% of US lawyers were using tools for their legal work.[136] But while the majority of respondents to our survey were women (78%), a much higher proportion of men than women reported using the tools in their work. (47% v. 17%)[137] While available data suggests that 55% of visits to the ChatGPT website are by men and 45% are by women,[138] male respondents to our survey were nearly three times as likely as their female counterparts to be using the tools.[139] Women's outlook regarding the future use of tools was also less optimistic—50% of women were unsure of or not planning to use AI tools in their work in the future as compared to 28% of men.[140]

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135. *See infra* Appendix, Demographics of Survey Takers.

136. LEXISNEXIS, INTERNATIONAL LEGAL GENERATIVE AI REPORT: DETAILED SURVEY FINDINGS 3 (2023), https://www.lexisnexis.com/pdf/lexisplus/international-legal-generative-ai-report.pdf [https://perma.cc/K2B4-YNYC]. *But see* Conf. Bd., *Survey: Majority of US Workers Are Already Using Generative AI Tools—But Company Policies Trail Behind*, PR NEWSWIRE (Sept. 13, 2023), https://www.prnewswire.com/news-releases/survey-majority-of-us-workers-are-already-using-generative-ai-toolsbut-company-policies-trail-behind-301925743.html [https://perma.cc/26JC-XYR2] (report of the Conference Board reporting that 56% of respondents to their survey, which included workers in general, not restricted to lawyers, were using generative AI for work tasks).

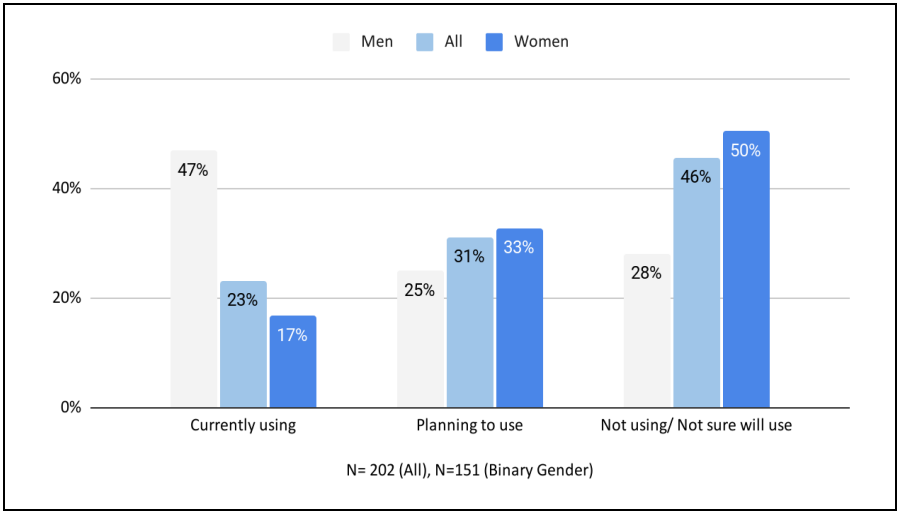
137. p = .001. *See infra* Appendix, Table A1.

138. *Chat.openai.com Website Traffic Demographics*, SIMILARWEB, https://www.similarweb.com/website/chat.openai.com/#demographics [https://perma.cc/L5L6-RYPN] (estimating that of visitors to chat.openai.com, that about 55% are male, and around 45% are female).

139. *See, e.g.*, SARAH DI TROIA ET AL., STANFORD INST. OF HUM.-CENTERED A.I., INSPIRING ACTION: IDENTIFYING THE SOCIAL SECTOR AI OPPORTUNITY GAP 22 (2024), https://hai.stanford.edu/sites/default/files/2024-02/Working-Paper-Inspiring-Action.pdf [https://perma.cc/MD69-6TXZ] (showing that, among 233 social sector professional respondents to a survey, male survey respondents were about twice as likely as female respondents to be using the technology daily and were about half as likely as to have never used generative AI tools).

140. p = .002. *See infra* Appendix, Table A1.

Figure 1: Current use of AI Tools by
Legal Aid Survey Respondents by Gender

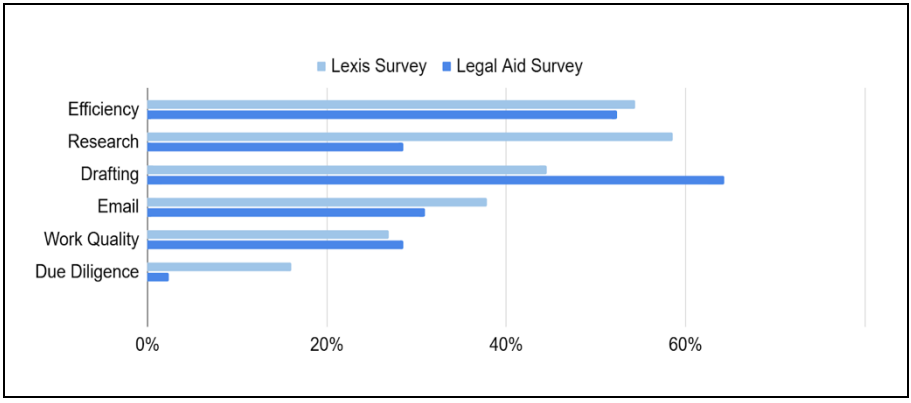


Legal aid professionals that affirmatively reported using generative AI tools[141] cited a wide range of uses, resembling those reported by corporate lawyers (Figure 1). But while both groups cited increasing efficiency and quality, drafting emails, and drafting documents as priority uses, the groups also diverged in how they used the tools. Corporate lawyers, for example, were much more likely to use generative AI for due diligence than were legal aid professionals, for example. Meanwhile, certain use cases popular among legal aid lawyers, like translation, were not offered as options to Lexis survey respondents. Consistent with their more extensive use of the tools, men reported a higher average number of ways of using AI than women.[142]

141. N = 43.

142. 1.5 v. 0.5 ways of using the tools (p = .001) When we adjusted for use, there were no gender differences in the mean number of use cases identified by men or women. *See infra* Appendix, Table A1.

Figure 2: Selected Generative AI Uses by Respondents to Lexis-Nexis and Legal Aid Surveys



Besides by gender, we did not see any significant differences in the current use of AI by any other particular group.[143] However, respondents with 25 or more years of experience were more likely to be using AI than those with 15–25 years of experience.[144] In addition, those providing direct service to clients reported significantly more ways of using AI than those in leadership roles.[145]

We also asked respondents to our initial baseline survey who had not used generative AI tools about the ways in which the tools potentially could be used.[146] In contrast to *actual* use, which varied significantly by gender, there were no statistical differences in the number of *potential* uses across profiles.[147] We asked respondents about their perceptions of the concerns and benefits associated with generative AI tools.[148] Across the board, respondents to our initial survey expressed

143. Specifically, by race/ethnicity (p = .317), years of experience (p = .250), age (p = .857), or role (p = .913).

144. 41% v. 16%, p = .077.

145. p = .018. There were no significant differences in the number of ways of using AI between different race/ethnic groups (p = .759), level of work experience (p = .211), and age when previous experience with AI was and was not taken into account.

146. *See infra* Survey 1, question 8.

147. *See infra* Appendix, Table A1.

148. *See infra* Survey 1, questions 8(b), 8(f), and 9.

that they believed AI to be generally beneficial[149] but also raised a number of concerns.[150]

B. Productivity, Attitudes, and Intentions to Continue Use

We next considered the experiences of people who participated in the pilot. Unlike earlier experimental simulations which used uniform tests or tasks created for purposes of the pilot described in Part II, our field trial design provided the actual conditions that people in the field would encounter when introduced to the tools. While this made it impossible for us to assess performance gains or losses with reference to a standardized task, we were still able to assess participant experiences with the tools in a few ways.

First, we directly asked users a set of questions about how their experiences and attitudes changed over the course of the pilot.[151] While this subjected our responses to the well-rehearsed challenges of self-reporting bias, it did provide a way for us to consider outcomes that were keyed to improvement relative to each participant's baseline. Our second method of assessment attempted to reduce the impact of reporting bias by looking for within-respondent differences to the same questions, provided just before and then just after the pilot. Though such a "quasi-experimental design" has its limitations, the short period of time between the two surveys (pre and post), with two to three months elapsing between surveys, reduced the chance that reported changes were attributable to secular factors rather than to exposure to tools—though we acknowledge the rapid evolution of AI tools during this period of time. We also asked participants about their intentions to continue use, including paid use, of the tools and which tools were favored. Finally, because we randomized an aspect of treatment—whether or not tools were rolled out with concierge support—we were able to assess the impact of this support, if any, on the outcomes and experiences of legal aid professionals.

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149. Q: Based on what you know about AI conversational tools like ChatGPT, how beneficial do you believe they are in assisting with tasks or answering questions? A: very beneficial = 5, beneficial = 4, somewhat beneficial = 3, neither beneficial nor harmful = 2, somewhat harmful = 1. (M = 3.4, SD = 1.0). *See infra* Survey 1, question 9.

150. Among 8 options, citing 3.7 concerns on average, with confidentiality, inaccuracy, data privacy, and ethical concerns ranking highest.

151. As reflected in questions that asked about comparative increases or decreases in usage (S2 Q8), satisfaction (S2 13), productivity (S2 12), and concerns (S2 13). *See infra* Survey 2, questions 4, 13–16 (survey questions asking about comparative increases or decreases in usage, satisfaction, productivity, and concerns).

Positive Experiences with Generative AI

Respondents generally reported increases in use and satisfaction with generative AI following exposure through the pilot. 86% reported more AI use during the pilot, and 63% reported greater satisfaction with AI after the pilot (Table 1). There were no significant differences with respect to changes in use or satisfaction with AI due to the pilot by gender, race, experience, or age. However, a higher percentage of respondents in leadership and direct service roles reported increased satisfaction than respondents in "other" roles.[152]

Table 1: Average self-reported changes in usage, productivity, or concerns as a result of the pilot[153]

Metric	Average Value (sd) / Share	Scale, Data Notes
Comparative Usage of AI	2.78 (0.6) / 86%	1=Less, 2= Same, 3=More / Share reporting some increase
Comparative Satisfaction	2.64 (0.5) / 63%	1=Less, 3= More / Share reporting some increase
Productivity Increase	90.2%[154]	Share reporting some increase
Level of Concern about AI	70.3%[155]	Share reporting same or more concern

sd = standard deviation

152. p = .030.

153. S2 Q8 (usage), S2 11 (satisfaction), S2 12 (productivity increase), S2 13 (more or less concerned). *See infra* Survey 2, questions 4, 13–15 (survey questions used for usage, satisfaction, productivity increase, and more or less concerned).

154. Breakdown: 9% no increase, 50% a little bit of an increase, 21% a medium increase, 14% a significant increase.

155. Breakdown: 12.5% more concerned, 58% just as concerned, 30% less concerned.

In terms of productivity, 90% reported some increase in productivity, with 25% reporting a medium or significant increase in productivity after the pilot (Table 1). While there were no significant differences with respect to productivity increases with AI use after the pilot by gender, race, years of legal experience, or age, there was a significant difference in productivity increase by role, with supervisors reporting higher levels of productivity gains than non-supervisory roles.[156]

In addition to considering participants' self-assessments as a result of their exposure, we compared respondents' answers to the same questions asked before and after the pilot. Before the trial, pilot participants reported an average of 1.2 uses for the tools. After the pilot, the average number of ways participants used the technology increased to 4.2, a significant difference.[157]

But even as respondents generally reported greater use and productivity with the tools, they reported similar levels of concerns about the tools. Over 70% of respondents reported being just as or more concerned by AI tools (Table 1). In addition, the mean number of concerns post-pilot was similar to the mean number of concerns pre-pilot, and there were no significant differences in the level of concern about AI after the pilot by demographic group.[158]

How did pilot participants balance higher levels of productivity with the persistent concerns they experienced based on the use of the tools? The responses to our question regarding participant intentions to continue using the tool provide one answer: 75% of participants in the pilot that responded to the second survey indicated their intention to keep using the tools, around 83% of whom indicated that they intended to continue use of paid tools[159] (Table 2). Among the tested tools, legal aid professionals favored ChatGPT, even over higher priced and specialized products (Figure 3).

156. $p = .033$.

157. $p < .001$ (1.2 (SD = 2.35) vs 4.2 (SD = 1.63)). Averages calculated among survey respondents that received the same wording of the question as it appeared in the exit survey.

158. Post-pilot: M = 4.7, SD = 2.2, v, pre-pilot: (M = 4.9, SD = 2.0) (ns). Averages calculated among survey respondents that received the same wording of the question as it appeared in the exit survey.

159. 83.3% of 75% = 62.5%. *See infra* Table 2.

Figure 3: Which Generative AI tool(s) do you plan to continue to use?

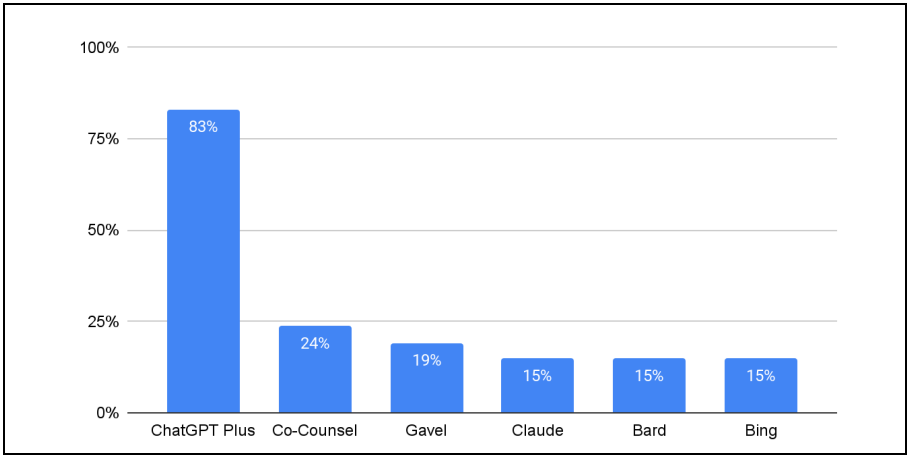

Table 2: Post-Pilot Outcomes

Question	Average Value (sd)	Scale, Data Notes
How often Used	3.14 (1.0)	1=None, 5=Everyday
Variety of ways Used	4.2 (1.6)	Selected from a list of 8 ways
Quality of AI output	1.70 (0.5)	1=Novice, 3=Expert
Positive Impact of AI on my job	71.4%	Share reporting that GenAI tools will have a positive impact
Future Use (any)	75%	Share reporting intent to use tools in the future
Future Paid Use (any)	62.5%	Share reporting intent to use paid tools in the future

Answers to other questions provided their rationale: on average, 64% of people rated the output of the tools to be at the level of an experienced or expert worker.[160] In addition, 71% of pilot participants reported a positive outlook with respect to the impact of AI on their work[161] (Table 2).

160. 3% expert, 33% novice. *Cf* Conf. Bd., *supra* note 136 (reporting responses of 45% = experienced, 31% = novice, 10% = expert (sic)).

161. Answering "a." to the Question: "Please check the box that best describes your outlook on the role of AI in your work." Answer: "a. AI will replace elements of my job in a positive way—

C. Differences by Gender Not Replicated Post-Pilot

While pilot participants generally reported having positive experiences, we sought to determine whether the gender differences in use we observed pre-pilot persisted in our post-pilot results. We first established baseline measurements of pilot participants that completed our initial survey. This self-selecting group had demographic traits that differed from the complete survey group, skewing younger and less experienced. It was also plausibl that they differed in other ways from survey respondents, such as harboring more optimistic and positive attitudes towards the tools.

Indeed, the pilot group featured a greater share of people who were using generative AI tools.[162] But we also observed a gender divide between pilot participants similar to what we saw among the overall group: a much higher share of men than women[163] reported use prior to the pilot.[164] In addition, there were also significant differences in the number of identified and prospective uses detected, suggesting latent demand and interest among the women who opted into the pilot.[165] Further, while pilot participants, like the overall population of survey-takers, generally felt that AI was beneficial pre-pilot,[166] male pilot participants found AI tools to be marginally more beneficial than female pilot participants.[167] Again, the difference disappeared when exposure was taken into account.

But, as was true of the earlier findings above about changes in use, satisfaction, productivity, and concern levels, we generally did not find any significant differences between the survey responses of men and women following the pilot.[168] This was true of the questions that were repeated regarding the use, attitudes, and concerns towards generative AI tools, as well as with newly introduced questions about

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e.g., by freeing up time for more valuable or creative tasks; b. AI will replace parts of my work in a negative way—e.g., by threatening my job altogether; c. I do not expect AI to replace any element of my job." (a: 71%, b: 26%, c: 3%). *See infra* Survey 2, question 17.

162. 21% vs. 35%.

163. 65% (men), 22% (women). *See infra* Appendix, Table A1.

164. $p = .012$.

165. For example, while men cited more ways of using the tool (* $p < .05$), women cited more ways that they would *consider* using the tools (* $p < .01$). Appendix Table A1: Differences in outcomes by gender in the pilot population before and after the pilot.

166. $M = 3.3$, $SD = 0.9$.

167. $p = .067$.

168. *See infra* Appendix, Table A1 (showing differences in outcomes by gender in the pilot population before and after the pilot).

the quality of work product and outlook regarding the role of AI and future intentions to use AI products. This suggests that once men and women were equally exposed to and provided with the tools, there was no statistically significant difference in their reported experiences.

D. Concierge Services Led to Better Outcomes

Our final hypothesis was that providing assistance along with paid generative AI tools could improve participant experiences and outcomes. As previously mentioned, it is equally easy to overestimate as it is to underestimate the power of generative AI tools.[169] Productivity gains can often only be accessed once initial frictions and learning curves are surmounted.

A comparison of outcomes by concierge v. standard group provides some support for our hypothesis. Reported usage during the pilot compared to before was slightly (but not significantly) higher in the standard group[170] than the concierge group.[171] But on every other metric, pilot participants who received concierge services reported better outcomes than those that did not. For five out of the nine outcomes, the differences were statistically significant. When adjusted for previous use, three out of the five outcomes remained significantly different for the concierge group than the standard group, implying that people with different levels of exposure to the tools could all benefit to some degree.[172] (See Appendix Table A2: Differences in outcomes by concierge or standard support during the pilot.)

IV. GENERATIVE AI USE CASES FOR LAWYERS AND LEGAL AID PROFESSIONALS

As noted above, three-quarters of pilot participants indicated they would continue using tools despite a similar share being as or more concerned by AI tools after the pilot.[173] While this finding at first may seem counterintuitive, a closer examination of the use cases reveals how participants were able to manage the risks and still leverage AI.

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169. Anton Korinek, *Generative AI for Economic Research: Use Cases and Implications for Economists,* 61 J. ECON. LITERATURE 1281, 1283 (2023).

170. 2.8 (Standard group) vs. 2.7 (concierge group).

171. Q: Compared to before the pilot, your usage of Generative AI tools during the pilot was A: 3 = More than before, 2 = the same as before, 1 = less than before.

172. *See infra Appendix*, Table A2 (showing differences in outcomes by concierge or standard support during the pilot).

173. *See supra* Part III.

Participants reported significant efficiency gains on a wide range of tasks, such as translating legal text into a more accessible form or language, document summarization or analysis, brainstorming and ideation, getting to a first draft for legal writing, drafting and editing non-legal writing, and carrying out associated non-legal or operations-related tasks.[174] As we detail below, by selecting lower-risk tasks and exercising supervision, pilot participants were able to capture gains in efficiency and discover useful AI applications they reported that they were likely to use again.

A. Risks of AI Tools

Before we discuss the use cases in detail, we briefly address the top concerns that pilot participants identified both before and after the pilot. These concerns are consistent with key AI risks and limitations discussed in the literature and emerging ethical guidance: confidentiality, data privacy, inaccurate results, and hallucinations.[175]

Data privacy and confidentiality ranked high among reported concerns before and after the pilot. As the California State Bar explains in its Practical Guidance on the Use of Generative AI, "[g]enerative AI products are able to utilize the information that is input, including prompts and uploaded documents or resources, to train the AI, and might also share the query with third parties or use it for other purposes.[176] "Even if the product does not utilize or share inputted information, it may lack reasonable or adequate security."[177] Thus, the State Bar advises lawyers using generative AI to take steps to ensure the confidentiality of client information is protected, such as using an AI system with stringent security and confidentiality protocols, anonymizing client information, and/or not entering identifying information.[178] As is reflected in the use case database, many pilot

174. *See* discussion *infra* Section IV.B.

175. *See supra* Part III. "Ethical concerns" also ranked high among concerns, but we do not address it separately since it overlaps with the other concerns that ranked highest.

176. STANDING COMM. ON PRO. RESP. & CONDUCT, STATE BAR OF CAL., PRACTICAL GUIDANCE FOR THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE IN THE PRACTICE OF LAW, https://www.calbar.ca.gov/Portals/0/documents/ethics/Generative-AI-Practical-Guidance.pdf [https://perma.cc/E54Z-7FCV].

177. *Id.* at 2.

178. *Id.* ("A lawyer must not input any confidential information of the client into any generative AI solution that lacks adequate confidentiality and security protections. A lawyer must anonymize client information and avoid entering details that can be used to identify the client. A lawyer or law firm should consult with IT professionals or cybersecurity experts to ensure that any AI system in which a lawyer would input confidential client information adheres to stringent security,

participants used the consumer-facing version of ChatGPT-4 but omitted any information that could be used to identify their clients.

Hallucinations are the confident generation by AI tools of *inaccurate results*, such as nonexistent case citations, inaccurate legal information, and incorrect information about court procedures or deadlines. In addition to the sanctions order in *Mata v. Avianca*[179] discussed above, national media outlets reported on former Trump lawyer Michael Cohen's use of Google Bard to unknowingly generate fake case citations, which his lawyer used in a motion.[180] Other lawyers have also been caught for submitting hallucinated, nonexistent case citations.[181] In response, courts in several jurisdictions have issued local rules or standing orders mandating the disclosure of the use of generative AI and/or a certification that the accuracy of any AI-generated content was verified.[182] The State Bar of California also

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confidentiality, and data retention protocols."); *see also* FLA. BAR, FLORIDA BAR ETHICS OPINION: OPINION 24-1 at 1–2 (2024), https://www-media.floridabar.org/uploads/2024/01/FL-Bar-Ethics-Op-24-1.pdf [https://perma.cc/V2L2-MQWU].

179. Mata v. Avianca, Inc., No. 22-CV-1461, 2023 WL 4114965, at *1 (S.D.N.Y. June 22, 2023).

180. Benjamin Weiser & Jonah E. Bromwich, *Michael Cohen Used Artificial Intelligence in Feeding Lawyer Bogus Cases*, N.Y. TIMES (Dec. 29, 2023), https://www.nytimes.com/2023/12/29/nyregion/michael-cohen-ai-fake-cases.html [https://perma.cc/5EMN-CTUH]; United States v. Cohen, No. 18-CR-602, 2023 WL 8635521, at *1 (S.D.N.Y. Dec. 12, 2023) (order to show cause requiring submission of declaration, including "a thorough explanation of how the motion came to cite cases that do not exist and what role, if any, Mr. Cohen played in drafting or reviewing the motion before it was filed").

181. *See, e.g.*, Park v. Kim, 91 F.4th 610, 615–616 (2d Cir. 2024) (finding that attorney, who submitted brief relying on non-existent case suggested by ChatGPT, failed to make the reasonable inquiry required by Rule 11 and referring attorney for further investigation and disciplinary proceedings); People v. Crabill, No. 23PDJ067, 2023 WL 8111898, at *1 (Colo. O.P.D.J. Nov. 22, 2023) (suspending attorney for failing to verify cases provided by ChatGPT); David Wagner, *This Prolific LA Eviction Law Firm Was Caught Faking Cases in Court. Did They Misuse AI?*, LAIST (Oct. 12, 2023, 5:00 AM), https://laist.com/news/housing-homelessness/dennis-block-chatgpt-artificial-intelligence-ai-eviction-court-los-angeles-lawyer-sanction-housing-tenant-landlord [https://perma.cc/GQC8-TVD8].

182. *See, e.g.*, U.S. DIST. CT. FOR E.D. TEX., LOCAL RULES CV-11(g), AT-3(m) (Dec. 1, 2023) (requiring lawyers and pro se litigants who use technology, such as ChatGPT, Bard, Bing, or generative AI services, to "review and verify any computer-generated content to ensure that it complies with" Rule 11 and all other applicable standards of practice); JUDGE PETER H. KANG, U.S. DIST. CT. FOR N. DISTR. OF CAL., STANDING ORDER FOR CIVIL CASES ¶ VII(C) (July 14, 2023) (outlining detailed guidance for AI and filings with the court, AI-generated evidence, and AI and confidentiality; and clarifying that the guidance does not apply to "use of traditional legal research, word processing, spellchecking, grammar checking, or formatting software tools (*e.g.,* Lexis, Westlaw, Microsoft Word, or Adobe Acrobat)"); JUDGE MICHAEL M. BAYLSON, U.S. DIST. CT. FOR E. DIST. OF PA., STANDING ORDER RE: ARTIFICIAL INTELLIGENCE (June 6, 2023); JUDGE BRANTLEY STARR, U.S. DIST. CT N.D. TEX, MANDATORY CERTIFICATION REGARDING GENERATIVE ARTIFICIAL INTELLIGENCE, https://www.txnd.uscourts.gov/judge/judge-brantley-starr [https://perma.cc/2L4G-K5CP]. Some commentators have expressed concerns that such rules or orders

advises lawyers that their existing professional obligations require them to verify and correct any AI-generated output, including any analysis and case citations, consistent with the duty of supervision.[183]

B. Beneficial Uses of AI Tools, By Type of Task

We asked pilot participants to submit helpful use cases (including prompts and outputs),[184] and classify each use case by the following types of task: (1) brainstorming and ideation, (2) document summarization or analysis, (3) translation, (4) legal research, (5) legal writing, (6) nonlegal writing, (7) automating letters, (8) intake processes, (9) development work (e.g., grant writing), and (10) other.[185] We also asked participants to rate each use case they submitted based on its usefulness (on a scale of 1 to 5),[186] likelihood they would reuse the use case (on a scale of 1 to 3),[187] and how much more efficient they were as a result of using AI for the task.[188]

As shown below and in the companion database (available at https://bit.ly/AIA2J), pilot participants reported efficiency gains on a wide range of tasks.[189] We discuss below a sampling of the submitted use cases and associated risks, time savings, and likelihood of repeat use.

1. Document Summarization or Analysis

Non-generative AI has been used for years to save time and increase efficiency in reviewing documents for purposes of eDiscovery, due diligence, and contract management.[190] Generative AI solutions build on those capabilities and offer the potential to review documents

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may be overbroad and/or will discourage appropriate uses of generative AI. *See, e.g.*, Maura R. Grossman et al., *Is Disclosure and Certification of the Use of Generative AI Really Necessary?*, 107 JUDICATURE, no 2., 2023, at 68, 69; *see* Choi et al., *supra* note 94 (manuscript at 46–47).

183. STANDING COMM. ON PRO. RESP. & CONDUCT, *supra* note 176, at 4; *see also* FLA. BAR, *supra* note 178.

184. We asked pilot participants not to submit or to redact any confidential client information, so there were some limits on the use cases we could receive. We also re-reviewed all of the entries to protect against privacy breaches. Chien & Kim, *supra* note 33.

185. *Id.*

186. The question asked for: "Rating you would assign to this use case (1–5) 1 = minimally if at all useful, 5 = extremely useful." *Id.*

187. The question asked: "How likely are you to reuse this use case 3 = very likely, 2 = somewhat likely, 1 = unlikely." *Id.*

188. The question asked: "Please estimate how much more efficient you were as a result of using AI." *Id.*

189. *See id*.

190. *See infra* Part I.

faster and at lower cost, as well as generate summaries or analyses related to the documents, such as timelines, case narratives, and contract negotiating points or redlines.[191] Even if the user provides an AI tool with documents to summarize, this task is not without risks. The AI tool may generate inaccurate or incomplete summaries by, for example, missing key nuances that an experienced lawyer may be more likely to catch.[192] Nonetheless, given the speed with which LLMs can process and analyze voluminous datasets, participants found document summarization use cases to increase their productivity consistent with empirical studies.[193]

Use Case: Summarize court opinions and records

Michael Semanchik, executive director at the Innocence Center, used CoCounsel to draft summaries of Court of Appeal opinions and trial records to help identify cases with strong evidence of innocence.[194] The document summarization feature in CoCounsel similarly could be used by others to enable the summarization of court filings, deposition and trial transcripts, court orders, and other lengthy documents.

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191. *See, e.g.*, *Data Solutions*, RELATIVITY, https://www.relativity.com/data-solutions/air/ [https://perma.cc/49TS-KSDN] ("From finding the most relevant documents to building your case narrative, aiR will lighten your workload every step of the way"); HARVEY, https://www.harvey.ai/ [https://perma.cc/D24G-FRQS]; Johnstone, *supra* note 4 (generative AI tool Harvey's uses include "document analysis, interpretation and summarisation, content creation and research").

192. Lauren Martin et al., *Better Call GPT*, Comparing Large Language Models Against Lawyers 12–13 (2024), https://arxiv.org/pdf/2401.16212.pdf [https://perma.cc/XQ2G-ASLA].

193. *See, e.g.*, *id*. at 10–11 (finding that LLMs performed "on par" with junior lawyers and legal processor outsourcers (LPOs) in determining legal issues within contracts, with the fastest LLMs completing the contract review in an average of 0.728 minutes in contrast to an average of 56.17 minutes for a junior lawyer and 201 minutes for an LPO).

194. *See* Presentation on Use of AI Tools, https://docs.google.com/presentation/d/16NE_yjXLKYzprlqo0dtGAJwWvupJwRwg/edit#slide=id [http://perma.cc/22A4-4AP8]; *see also* Bernard Marr, *How Generative AI Is Used to Fight Miscarriages of Justice at the California Innocence Project*, FORBES (Oct. 6, 2023, 2:49 AM), https://www.forbes.com/sites/bernardmarr/2023/10/06/how-generative-ai-is-used-to-fight-miscarriages-of-justice-at-the-california-innocence-project/ [https://perma.cc/KX88-48WB] (deploying CoCounsel to "assist with time-consuming and highly technical work"); Matt Reynolds, *California Innocence Project Harnesses Generative AI for Work to Free Wrongfully Convicted*, AM. BAR ASS'N J. (Aug. 14, 2023, 8:45 AM), https://www.abajournal.com/web/article/california-innocence-project-harnesses-generative-ai-for-work-to-free-wrongfully-convicted [https://perma.cc/KR6H-UPTC] (utilizing CoCounsel to narrow focus on commonly seen issues in wrongful conviction cases).

Use Case: Summarize and rewrite website text

A director of community engagement[195] asked ChatGPT-4 to summarize and rewrite the history webpage for her legal services organization.[196] Referring to the use case as a "[h]uge time saver," the participant reported that it was extremely useful (5 rating) and resulted in 80% time savings.[197]

> [Prompt #1:] Summarize the text below in 3–5 paragraphs, in plain language, and focus on the early history of inland counties legal services while omitting extraneous details. The resulting output will be for a nonprofit legal aid organization's history page and should illustrate the focus on ensuring access to justice. [*1878-word text of webpage omitted.*]
>
> . . . .
>
> [Prompt #2:] Alter the text above to deemphasize Legal Aid Society of San Bernardino and focus only on the formation and evolution of Inland Counties Legal Services.
>
> [ChatGPT-4 Output #2:] The history of Inland Counties Legal Services (ICLS) begins in the late 1950s with the establishment of the Legal Aid Society of Riverside (LASR). Founded in 1958, LASR was created to provide legal assistance to the needy and to collaborate with courts, the Bar, and social agencies in the administration of justice.[198]

2. Brainstorming and Ideation

Generative AI can help stimulate thinking and generate ideas, arguments, and counterarguments. As demonstrated by *Mata* and the Cohen case discussed above, there is a significant risk that the AI may generate irrelevant, biased, or inaccurate ideas, including nonexistent case citations and arguments that do not support the client's best interests.[199] While this risk may be mitigated by grounding the LLM's

195. Tori Praul, LINKEDIN, https://www.linkedin.com/in/toripraul/ [https://perma.cc/ZTD4-DGQH].

196. Chien & Kim, *supra* note 33, no. 66.

197. *Id.*

198. The full chat for this use case is available at Chien & Kim, supra note 33, no. 66.

199. Mata v. Avianca, Inc., No. 22-CV-1461, 2023 WL 4114965 (S.D.N.Y. June 22, 2023); United States v. Cohen, No. 18-CR-602, 2023 WL 8635521 (S.D.N.Y. Dec. 12, 2023).

brainstorming and ideation on user-provided data, such as a complaint or proposed legislation, the risk is not entirely eliminated.

Use Case: Brainstorming discovery requests

A pilot participant used ChatGPT-4 to brainstorm discovery requests.[200] The prompt, shown below, does not upload a copy of the complaint, but ChatGPT generated ten interrogatories and ten requests for production.[201] The user associated this use case with a 40% increase in efficiency, rating it as useful (3 of 5) and somewhat likely to be reused (2 of 3).[202]

> [Prompt:] I am an attorney litigating a lawsuit against the State of Louisiana, challenging the state's policy and practice of requiring persons with felony convictions to produce documentary proof of eligibility in order to register to vote. [] Can you give me ideas for interrogatories and requests for production?
>
> [ChatGPT-4:]
>
> . . . .
>
> ### Interrogatories
>
> 1. **Policy Details**: Please describe in detail the policy requiring persons with felony convictions to produce documentary proof of eligibility in order to register to vote, including any and all amendments and revisions made to this policy since its inception.
>
> 2. **Implementation Procedures**: Describe the procedures followed by the Secretary of State's office in implementing and enforcing the aforementioned policy.
>
> . . . .
>
> ### Requests for Production
>
> 1. **Policy Documents**: All documents pertaining to the development, implementation, revision, and enforcement of the policy in question.

200. Chien & Kim, *supra* note 33, no. 24.
201. *Id.*, no. 24.
202. *Id.*

> 2. **Statistical Data**: Any statistical data or analyses related to the impact of this policy on voter registration and turnout, particularly among individuals with felony convictions.
>
> 3. **Training Materials**: Copies of training materials and guidelines given to employees involved in voter registration under this policy.
>
> . . . .
>
> Remember, these are just starting points, and you may need to tailor these to the specifics of your case and legal strategy.[203]

Notably, ChatGPT-4 reminded the user that "these are just starting points" and should be tailored.[204] If the user had uploaded the complaint and relevant rules or orders, the AI tool may have generated more tailored discovery requests.

3. Legal Research

Pilot participants submitted fewer legal research use cases, and we saw significant variation in the quality and usefulness based on participants' comments. Notwithstanding ChatGPT's limitation as a general-purpose tool, some participants found beneficial ways to use it for legal research use cases for preliminary research or to confirm the accuracy of research that they had already performed using traditional research methods.

Use Case: Quick research during legal clinic

A pilot participant used ChatGPT-4 to perform quick research while volunteering at a Workers' Rights Clinics at Legal Aid Work.[205] The clinic offers a limited-scope engagement where the attorney may have a very brief period to conduct necessary research before providing the client with basic legal information and/or referrals at the end of the appointment.[206] The attorney associated the legal research use

203. Chien & Kim, *supra* note 33, no. 24.
204. *Id.*
205. *Id.*, no. 18.
206. *Workers' Rights Clinic*, OPENAI, https://legalaidatwork.org/clinics/workers-rights-clinic/ [https://perma.cc/S93H-N89E].

case with a 40% increase in efficiency, rating it as extremely useful and very likely to be reused.[207]

> [Prompt:] [W]hat are the eligibility rules for California state disability insurance
>
> [ChatGPT-4:] California State Disability Insurance (SDI) provides short-term benefits to eligible workers who have lost income due to a non-work-related illness, injury, or pregnancy. To be eligible for SDI, you must meet the following criteria:
>
> - Paid into SDI: You must have had SDI deductions from your paycheck. Most California employees contribute to this program through payroll deductions.
> - Minimum Earnings: You must have earned at least a minimum amount in wages during a 12-month base period. This base period is the 12 months prior to the start of your disability.
>
> . . . .
>
> It's important to note that specific situations and exceptions may apply, so it is always a good idea to check the most current eligibility criteria from the California Employment Development Department (EDD), which administers the SDI program.[208]

Use Case: Confirmatory legal research

A pilot participant asked CoCounsel to conduct asylum-related research after having already done some research using more traditional means. CoCounsel provided a short memo, which correctly concluded there was no clear case law. The user reported that the use case resulted in a 20% increase in efficiency, rating it as useful (3 of 5) and very likely to be reused (3 of 3).[209]

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207. Chien & Kim, *supra* note 33, no. 18. Mr. Clune also submitted and ranked as extremely useful another use case requesting an overview of liquidated damages in wage claims. *Id.*, no. 11.

208. *Id.*, no. 18.

209. *Id.*, no. 23.

4. Legal Writing

Since large language models are trained on massive volumes of text, pilot participants found ways to use the AI tools as writing assistants. With respect to legal writing, there is a significant risk that the AI may generate inaccurate or biased content.[210] As a result, we found that pilot participants generally tended to find their most useful AI writing use cases on letters or other writing that did not require citations to cases or other legal authority.

Use Case: Draft an Agreement

A pilot participant asked ChatGPT-4 to draft a memorandum of understanding between two non-profits. The user rated the use case as extremely useful and very likely to be reused. Citing a 60% increase in efficiency, the user noted that they used the tool as a "jumping off point" to get a "good first draft."[211]

> [Prompt:] Draft a MOU between two non-profits where one non-profit, NAME1, agrees to administer a $75,000 emergency housing fund for the benefit of NAME2's clients. The term is 1/1/2024 to 12/31/2024
>
> . . . .
>
> [Clarifying prompt:] specific for CA?
>
> [ChatGPT-4:] MEMORANDUM OF UNDERSTANDING (MOU)
>
> . . . .
>
> 1. ADMINISTRATION OF EMERGENCY HOUSING FUND:
>
> NAME1 agrees to administer the Emergency Housing Fund in accordance with the following terms and conditions:
>
> 1.1. NAME1 shall establish and maintain a separate account specifically designated for the Fund in compliance with California state regulations.

210. *The key legal issues relating to the use, acquisition, and development of AI*, Thomson Reuters (Mar. 1, 2024) https://legal.thomsonreuters.com/blog/the-key-legal-issues-with-gen-ai/ [http://perma.cc7DG5-BGWB].

211. Chien & Kim, supra note 33, no. 50.

> 1.2. NAME1 shall accept and process applications for emergency housing assistance from NAME2's eligible clients during the term of this MOU, adhering to California law.
>
> 1.3. NAME1 shall review and evaluate all applications received from NAME2's clients to determine eligibility for emergency housing assistance based on predetermined criteria and in accordance with California guidelines.
>
> . . . .
>
> 2. NAME2 RESPONSIBILITIES: . . . .
>
> 3. CONFIDENTIALITY: . . . .
>
> 4. AMENDMENTS: . . . .
>
> 5. TERMINATION: . . . .
>
> 6. GOVERNING LAW: . . . .[212]

Use Case: Cease and desist letter

A pilot participant used ChatGPT-4 to write a cease-and-desist letter to a prospective client who had been abusive. The user fine-tuned the prompt: the fourth prompt and output are below. The user reported that the use case resulted in a 60% increase in efficiency, rating it as extremely useful and very likely to be reused.[213]

> [Prompt:] [W]rite a letter to a prospective client of our pro bono legal services agency who has been abusive toward the lawyer referral services agency that refers clients to our agency as well as to private attorneys. In the letter clarify that we have never represented the client and also request that the client cease contacting our agency and staff.
>
> [ChatGPT-4:]
>
> . . . .
>
> Re: Notice of Non-Engagement and Request to Cease Contact
>
> Dear [Prospective Client's Name],

212. *Id.*
213. *Id.*, no. 29.

> I am writing on behalf of [Your Pro Bono Legal Services Agency's Name]. Our agency has been informed by the lawyer referral service, which is a key partner in referring clients to both our agency and private attorneys, about their recent interactions with you. It is our understanding that these interactions have involved conduct on your part that is regarded as abusive [detail specific nature of the conduct if appropriate and legally advisable].
>
> We wish to clarify that [Your Agency's Name] has not entered into an attorney-client relationship with you and will not be able to provide you with legal representation. Our decision is based on our commitment to maintaining a safe, respectful, and professional environment for our staff and the people we serve.
>
> In addition, we kindly request that you refrain from any further contact with our agency and our staff. Continued communication from you will not be responded to, and any attempts at contact will be documented.[214]

Use Case: Draft letter to landlord

An attorney at Mental Health Advocacy Services used ChatGPT-4 to draft a letter to a landlord requesting a reasonable accommodation for a client who was about to be evicted. Noting that the eviction notice was rescinded, the attorney reported that the use case resulted in a 40% increase in efficiency, rating it as extremely useful and very likely to be reused.[215]

> [ChatGPT-4:]
>
> . . . .
>
> Subject: Understanding Schizophrenia—A Request for Accommodation
>
> Dear [Property Manager's Name],
>
> I am writing to you regarding [Tenant's Name], who is a resident in your property and is currently under my care for the treatment of schizophrenia. I feel it's important to provide

214. *Id.*
215. *Id.*, no. 3.

> you with some insights into schizophrenia to ensure a supportive living environment and to discuss reasonable accommodations that might be necessary due to the nature of this condition.
>
> . . . .
>
> In light of this, we kindly request that you consider the following reasonable accommodations, in accordance with the Fair Housing Act:
>
> - Flexible Communication: During periods of crisis, alternative methods of communication may be necessary.
> - Guest Policy Leniency: Allowing a support person to stay with the tenant during a crisis.[216]

5. Nonlegal Writing

A popular use case among pilot participants was nonlegal writing, including drafting emails, letters, requests for proposal, website or social media content, or other administrative documents. The prevalence of this use case may be related to the fact that many lawyers in the pilot group hold leadership positions at their organization.[217]

Use Case: Draft policy talking points

A Deputy Director at the Center for Gender & Refugee Studies[218] used ChatGPT-4 to generate talking points on why saving asylum protections is important. She commented that it was "a surprisingly excellent summary of some of the main arguments we make to policymakers and other stakeholders (including donors) for why our work around asylum law is important to support." She found no evidence of bias or hallucinations and thought it could serve as a "starter" for a first draft of a fundraising appeal.[219]

> [Prompt:] Help me explain why saving asylum in the United States is important

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216. *Id.*
217. *See infra* Appendix, Survey Demographics.
218. Chien & Kim, *supra* note 33, no. 56.
219. *Id.* She rated this use case as a 4 of 5 for usefulness, 2 of 3 for somewhat likely to use, and 40% time savings. *Id.*

> [ChatGPT-4:] Saving asylum in the United States is crucial for various ethical, legal, and humanitarian reasons. Here are some key points to help you explain the importance of preserving asylum:
>
> **Humanitarian Obligation**: Asylum is a fundamental human right established by international law to protect individuals who are fleeing persecution, violence, or other forms of severe harm in their home countries. By granting asylum, the United States demonstrates its commitment to upholding human rights and providing a safe haven for those in need.
>
> **Protection of Vulnerable Populations**: Asylum serves as a lifeline for individuals, including women, children, and minorities, who are particularly vulnerable to persecution, discrimination, or violence in their home countries. . . .
>
> **Global Leadership**: By maintaining a robust asylum system, the United States sets an example for other nations to prioritize the protection of refugees and asylum seekers.[220]

Use Case: Draft a social media post

A vice president at Legal Aid at Work asked ChatGPT-4 to draft a tweet based on a press release.[221] After receiving an initial draft, he followed up twice with more instructions. He rated the use case as extremely useful and very likely to be reused, and he indicated that the task resulted in an 80% gain in efficiency, i.e., the task took him about 20% as much time as it usually did.[222]

> [Prompt #1:] can you draft a tweet about this: . . . A lawsuit against the Department of Defense seeks to remove any mention of sexual orientation or homosexual conduct from the military records of discharged veterans. . . . [*Rest of ~1370 word press release omitted*]
>
> . . . .
>
> [Prompt #2:] revise to indicate that it's our lawsuit (Us being Legal Aid at Work and the Impact Fund)

220. *Id.*
221. *Id.*, no. 17.
222. *Id.*

. . . .

> [Prompt #3:] revise to indicate we're not announcing the lawsuit--it is ongoing--but that 30,000 veterans deserve better
>
> [ChatGPT-4 Output #3:] Certainly, here's a revised version of the tweet to reflect the ongoing nature of the lawsuit and the focus on the 30,000 veterans:
>
> 🏳️🌈⚖️ In our ongoing lawsuit with the Impact Fund, we stand for the 30,000+ veterans unjustly affected by "Don't Ask, Don't Tell." This fight is about more than records; it's about honoring every service member's dignity and contribution. Let's right these wrongs together. #JusticeForVeterans #LGBTQService #LegalAidAtWork
>
> 📎 [Link to the full article] . . . [223]

6. Client Intake and Automating Documents

As discussed above, AI and automation technologies can be used to automate the client intake process and the drafting of letters, court forms, and other documents.[224] Some no-code platforms, including Gavel, are leveraging generative AI to make it easier for users to automate their templates into automated workflows.[225]

Use Case: Generate workflow for durable power of attorney

The Director of Pro Bono Programs at Legal Aid of North Carolina reported using Gavel to convert a Durable Power of Attorney template document into a fillable workflow partially depicted below.[226] After the volunteer attorney inputs responses to each question, Gavel generates a draft document for review and verification by the attorney and client. The organization's goal is "to use Gavel to make inexperienced volunteer attorneys more comfortable completing these documents for clients and reduce the number of typos in volunteer work."

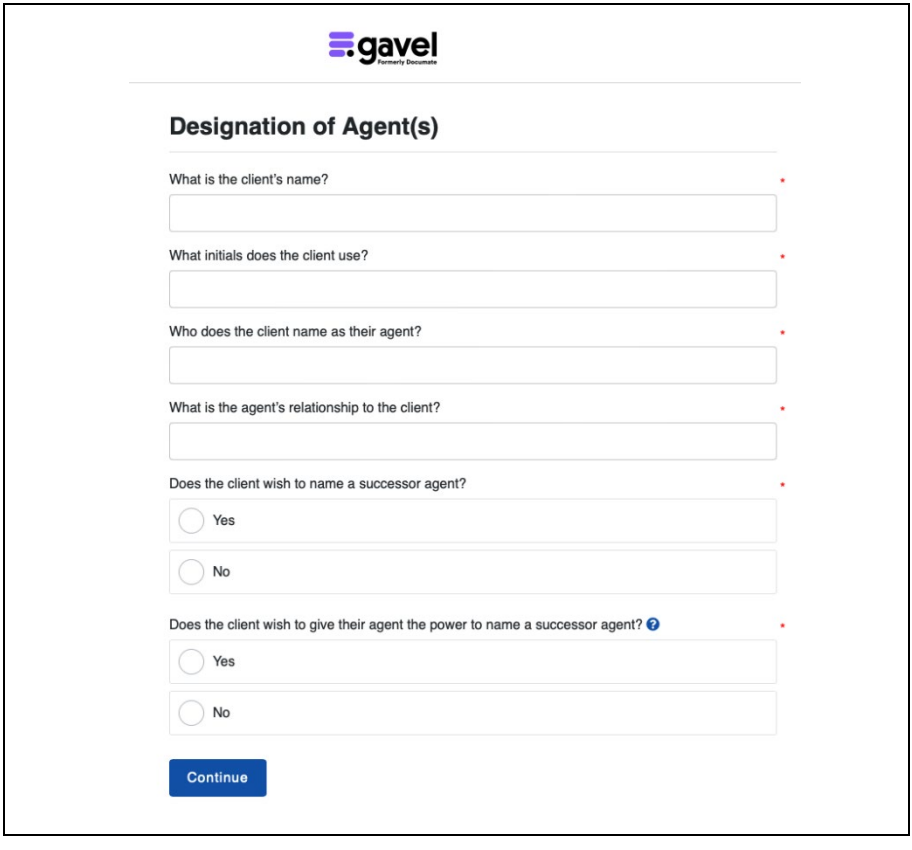
223. *Id.* For all the prompts and outputs for this use case, see 17-Copy of Use Case.

224. *See supra* Part II.

225. *Gavel Blueprint*, GAVEL, https://www.gavel.io/use-cases/gavel-blueprint [https://perma.cc/5CC7-EVZR].

226. Chien & Kim, *supra* note 33, no. 33. The final workflow is available at https://legalaidnc.gavel.io/run/playground2/POA/#/1.

The Director reported that the use case allowed her to save 80% time, rating it as extremely useful and very likely to be reused.[227]



7. Development Work (e.g., Grant Writing)

Legal aid organizations often lack the resources to serve everyone who comes to their doors for help. In response to a survey question about what it would take to double their impact, many respondents noted a need for more funding.[228] The development work use case would help address the funding needs by leveraging a number of the AI use cases discussed above, such as document summarization (e.g., analyzing current application requirements and past grant applications), brainstorming (e.g., generating ideas for a project plan or other

227. *Id.*

228. Survey responses detailed the use of funds to provide, e.g., "a coordinated intake system for the 60 plus immigration legal services who provide free legal services to low-income Texans," staff, advertising, and efficiency improvements.

component of the application), and nonlegal writing (e.g., drafting and editing the application).

Use Case: Draft grant application

Jane Ribadeneyra, Legal Services Corporation,[229] provided a detailed prompt to prepare a grant application using Claude, a general-purpose generative AI tool. Her presentation slides include excerpts of the draft application, which was generated within seconds.[230]

> I am going to give you a detailed APPLICATION GUIDE for a Technology Improvement Project (TIP) grant from the Legal Services Corporation and then I'm going to give you a TECHNOLOGY PLAN for a legal aid non-profit organization. Acting as a professional grant writer for the legal aid non-profit organization, follow the series of prompts below create a grant application for a Technology Improvement Project.
>
> [Prompt 1:] Following the Application Instructions in the APPLICATION GUIDE and using information about the current state of the legal aid organization's technology infrastructure in the TECHNOLOGY PLAN, provide three to four paragraph responses for each application question.
>
> [Prompt 2:] Write a PROJECT PLAN for the grant project, providing a succinct project goal, three objectives and three to five milestones that are align with the examples provided in the APPLICATION GUIDE.
>
> [Claude:] Here are 3–4 paragraph responses for each application question, based on the Application Guide and Technology Plan: . . . .[231]

8. Translation

Approximately 6.4 million Californians speak English "less than very well," and 17.2% of Californians cannot access the court system

229. Jane Ribadeneyra, LINKEDIN, https://www.linkedin.com/in/jribadeneyra/ [https://perma.cc/42X2-JV6A].

230. Chien & Kim, *supra* note 33, no. 81.

231. *Id.*

without language assistance.[232] Generative AI can help prepare translations, thereby making legal information and advice more accessible to individuals who do not speak English or have limited English proficiency.

Use Case: Translate client communication

A pilot participant asked ChatGPT-4 to translate a client communication using a simple prompt: "Please translate the following text into Spanish."[233] The user reported that it was "not a super great translation" and that it would need to be revised. Despite the need to "always double check the translation," the user reported that the translation use case resulted in a 20% increase in efficiency, rating it as very useful (4 of 5) and somewhat likely to be reused (2 of 3).[234]

Use Case: Translate legalese for nonlawyers

Some English-speaking and limited English-proficient individuals may benefit from an AI-generated translation of legalese into simple terms. This sample use case was used to explain a special needs trust to a client's friend or family member. The user associated it with a 40% increase in efficiency, rating it as extremely useful and somewhat likely to be reused.[235]

> [Prompt not provided]
>
> [ChatGPT-4:] Explaining a trust to a layperson requires breaking down the concept into simple terms. Here's a simplified explanation of a trust:
>
> A trust is like a special container that you can create to hold and manage your assets (like money, property, or investments) for a specific purpose or for the benefit of certain people. Imagine it as a safe or a box that you fill with your valuable items.
>
> There are typically three main roles in a trust:

232. JUD. COUNS. OF CAL., FACT SHEET: LANGUAGE ACCESS IMPLEMENTATION 1 (2023), https://www.courts.ca.gov/documents/LAP-Fact-Sheet.pdf [https://perma.cc/5WKA-8RGE] (citing U.S. Census Bureau, 2017–2021 American Community Survey 5-Year Estimates).

233. Chien & Kim, *supra* note 33, no. 45.

234. *Id.*

235. *Id.* no. 2.

- Grantor or Settlor: This is the person who creates the trust. Think of them as the owner of the box. They decide what goes into the trust and what rules should be followed.
- Trustee: The trustee is like the caretaker of the box. They are responsible for managing and safeguarding the assets in the trust according to the rules set by the grantor. Their job is to make sure the assets are used for the benefit of the beneficiaries . . . .

Keep in mind that there are various types of trusts, and the specifics can vary, but this simple explanation should provide a basic understanding for a layperson.[236]

9. Building Custom GPTs

During the course of the pilot, OpenAI launched custom GPTs.[237] We provided the concierge group with training on building custom GPTs during office hours. Several participants and researchers involved in the pilot built GPTs to explore the possibility of distributing legal information to volunteers and/or consumers at scale. The exploratory GPTs direct individuals to TexasLawHelp.org re-sources[238] and information on landlord-tenant law and eviction actions,[239] criminal expungement eligibility,[240] and special immigration juvenile petitions.[241] The CIO of Legal Aid Chicago also built a GPT to help write technology surveys for users at their organization.[242]

236. *Id.*

237. *List of Publicly Accessible Chat GPTs*, OPENAI, https://chat.openai.com/gpts [https://perma.cc/4WLY-FKUD].

238. *TexasLawHelp Navigator*, OPENAI, https://chat.openai.com/g/g-iHB5j3DzB-texaslawhelp-navigator [https://perma.cc/U5DU-RYEA]; Chien & Kim, *supra* note 33, no. 41.

239. Miriam Kim, *Arizona Eviction Bot*, OPENAI (Dec. 2023), https://chat.openai.com/g/g-7J13wp6qh-arizona-eviction-bot [http:/perma.cc/W9CB-LUSY]; *see also* Chien & Kim, *supra* note 33, no. 62.

240. *Arizona Expungement Help*, OPENAI, https://chat.openai.com/g/g-zKSocDTTs-arizona-expungement-help [https://perma.cc/73DA-U63M]; Rohit Rathish, *Maryland—Brute Force*, https://chat.openai.com/g/g-mDNjoP1OS-maryland-brute-force [http://perma.cc/3QPK-T66S].

241. Miriam Kim, *California SIJS Guide*, OPENAI, https://chat.openai.com/g/g-rKWt2oaMP-california-sijs-guide [https://perma.cc/J4CC-BQ7S].

242. *Tech Survey Assistant*, OPENAI, https://chat.openai.com/g/g-wfGZMqaB2-tech-survey-assistant [http://perma.cc/27UK-L93F]; *see* Chien & Kim, *supra* note 33, no. 27.

10. Other Tasks

Some of the "Other" use cases that did not clearly fall within another category included coding, database administration, and HR. We also note that over thirty of the beneficial use cases submitted through the pilot are administrative, legal operations, or management tasks, such as hiring,[243] payroll,[244] benefits,[245] training,[246] and procurement.[247]

11. Beneficial Use Cases

As discussed further below, across the uses in our companion case database, pilot participants reported an increase in efficiency of an average of 50% on a wide range of tasks. The following five categories of lower-risk use cases were popular among the pilot group and may therefore provide promising starting points for lawyers seeking to use generative AI in their work:

a. Lower-risk tasks where user-provided data or inputs limit hallucination risk

Pilot participants found beneficial use cases for lower-risk tasks where the input data grounds the AI model and limits the risk of hallucinations. This includes document summaries and brainstorming and ideation. Pilot participants asked ChatGPT-4 and CoCounsel to summarize, analyze, and, in some cases, rewrite a variety of documents ranging from court opinions[248] and expert declarations[249] to an organization's website that needs to be revised.[250] On the

243. Chien & Kim, *supra* note 33, no. 55 (drafting email to unsuccessful job applicants).

244. *Id.* no. 37 (drafting email to staff regarding payroll).

245. *Id.* no. 60 (drafting annual management process to review insurance coverage and rating the use case extremely useful, very likely to reuse, and 60% efficiency gain); *id.* ("I believe strongly the work product was much better/richer than if I attempted to do the task from scratch.").

246. *Id.* no. 7 (drafting training questions on housing law).

247. *Id.* no. 26 (writing a request for proposal). Submitted by Vivian Hessel, Legal Aid Chicago. *Leadership,* LEGAL AID CHI., https://legalaidchicago.org/who-we-are/our-people/leadership/ [perma.cc/UX8J-HJG4].

248. *See* Presentation on Use of AI Tools, *supra* note 194; *see also* Marr, *supra* note 194 (deploying AI to assist with time-consuming and highly technical work of reviewing case law); Reynolds, *supra* note 194 (using CoCounsel for legal research).

249. Chien & Kim, *supra* note 33, no. 34.

250. *See id.* no. 66 (referring to the use case as a "[h]uge time saver," the participant reported that it was extremely useful (5 rating) and resulted in 80% time savings). Submitted by Tori Praul, Deputy Director of Community Engagement, at Inland Counties Legal Services (ICLS). Tori Praul, *supra* note 195.

brainstorming end, brainstorming discovery requests[251] proved more useful and productive than the more complex task of brainstorming appellate arguments.[252]

b. Legal Writing Assistant—getting to a first draft faster

Pilot participants also found some beneficial ways to use generative AI to help them get to a first draft of legal writing faster, particularly where legal citations are not required. There were no use cases submitted for brief writing or other complex legal writing. Instead, pilot participants found increases in productivity through the use of AI to write and edit legal correspondence (e.g., letter to a landlord[253] or a cease-and-desist letter[254]), automate letters and legal documents (e.g., durable power of attorney[255]), and draft a contract.[256]

c. Writing assistant—getting the tone right

Several pilot participants shared that ChatGPT-4 is particularly useful as an assistant for writing or editing emails to ensure the right tone is used or to address difficult communications.[257] ChatGPT-4's simplification of legalese was also rated as extremely useful (e.g., explaining a trust).[258]

d. Preliminary or confirmatory legal research

Although we saw significant variation in the quality of the legal research use cases submitted, there were beneficial uses for preliminary or confirmatory research. For example, while ChatGPT-4 did fairly well on preliminary legal research on basic topics,[259] it struggled with more complex legal research questions.[260] This variation based

251. Chien & Kim, *supra* note 33, no. 24.
252. *Id*. no. 25.
253. *Id.* no. 83.
254. *Id.* no. 29.
255. *Id.* no. 33.
256. *Id*. no. 50.
257. *See, e.g.*, *id.* no. 22 (editing email to appear more professional); *id.* no. 55 (writing email to job applicants who did not get offer).
258. *Id.* no. 2.
259. *See, e.g.*, *id.* no. 46 (Prompt: "What are the legal requirements to be granted asylum in the US") (rated useful [3 of 5], likely to be reused [2 of 3], and described as "[g]ood for a quick overview/brief understanding, but would definitely want to double check all info with an expert source").
260. *See, e.g.*, *id.* no. 39 (Prompt: "What takes precedence, a circuit court of appeals decision or a subsequently issued board of immigration appeals decision?") (rated unlikely to be reused and

on complexity is not surprising given that ChatGPT-4 is not specially trained for legal use cases.[261] Notwithstanding ChatGPT's limitations as a general-purpose tool and the risk of hallucinations, some participants found beneficial ways to use it for legal research use cases for preliminary research[262] or confirmatory research (i.e., confirming the accuracy of research that they had already performed using traditional research methods).[263] Quick re-search in the middle of a legal aid clinic was also cited as extremely useful.[264]

e. Lower risk tasks that are nonlegal and have lower stakes

Pilot participants, who skewed heavily toward leadership positions (47.6%),[265] found very beneficial use cases in operations, administrative, or management tasks with lower stakes and minimal to no consequences for inaccuracy.[266] This includes nonlegal tasks like drafting internal staff emails,[267] social media content,[268] policy talking points,[269] policies,[270] and administrative planning documents.[271]

V. OBSERVATIONS AND RECOMMENDATIONS

In the following sections, we leverage our study's findings to propose ways in which generative AI can improve legal aid services, thereby advancing access to justice. Although this approach represents only one path to narrowing the justice gap, it is significant due to the central role that lawyers have in providing legal help to underserved populations. Nonetheless, the recommendations provided should be

did not save time); *id.* no. 40 (Prompt: "What takes precedence, a circuit court of appeals decision or a subsequently issued regulation?") (rated unlikely to be reused and did not save time).

261. *See* Jonathan H. Choi et al., *supra* note 92, at 387.

262. Chien & Kim, *supra* note 33, no. 11 (requesting an overview of liquidated damages in wage claims and rated as extremely useful, very likely to be reused, and resulting in an 80% increase in efficiency). Submitted by Kevin Clune, Legal Aid at Work. *Kevin Clune*, LEGAL AID WORK, https://legalaidatwork.org/staff-members/kevin-clune/ [https://perma.cc/YU9R-3HMU].

263. *See* Chien & Kim, *supra* note 33, no. 23.

264. *Id.* no. 18 (requesting information on eligibility for state disability insurance during clinic and rated as extremely useful, very likely to be reused, and resulting in a 40% increase in efficiency) Submitted by Kevin Clune, Legal Aid at Work. *Kevin Clune*, *supra* note 262.

265. *See infra* Appendix for demographic details.

266. *See supra* Section IV.B. Beneficial Uses of AI Tools, By Type of Task.

267. Chien & Kim, *supra* note 33, no. 59.

268. *Id.* no. 17.

269. *Id.* no. 56.

270. *Id.* no. 52.

271. *Id.* no. 31.

viewed as preliminary, due to the limited scope of our survey and pilot study, warranting further research, inquiry, and refinement.

A. Ensuring Equitable Access and Uptake of Generative AI Tools in Legal Aid

As discussed, the majority of legal aid attorneys in our pilot study reported that the generative AI tools provided were useful and advantageous. However, it is clear that the benefits of generative AI are available only to the extent that users access and utilize these technologies. Our findings indicate that simply making generative AI tools "available" does not guarantee equitable or optimal outcomes.

For example, our survey showed stark gender differences in the use of generative AI tools: female respondents to our survey were only about a third as likely as their male counterparts to be using them.[272] While we may be among the first to document a gender divide in the uptake of AI tools, differences among men and women with respect to access and attitudes towards technology have long been observed.[273] In 2020, as part of its 2020–2025 "Gender Equality Strategy," the European Union published a report that documented and explained the lower rate of confidence and trust, and higher levels of concern about digital technologies, including robots. These concerns traced, in turn, back to the gendered design of digital technologies and lack of testing of technologies with women.[274]

The pronounced gender disparity in the adoption of AI tools is particularly concerning within the legal aid sector, where women represent over 75% of the workforce, even though they remain underrepresented among practicing attorneys at large.[275] Women also

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272. *See infra* Appendix, Table A1.

273. *See e.g.*, Zhihui Cai et al., *Gender and Attitudes Toward Technology Use: A Meta-Analysis,* 105 COMPUTS. & EDUC. 1, 2–10 (2017) (providing a meta-analysis of studies of gender differences in attitudes toward technology and finding the gaps to have reduced along certain dimensions (affect and self-efficacy) but not along others (belief)). A relative recent review of survey and related empirical evidence on gender gaps in digital technologies, from a global perspective, is provided by ALISON GILLWALD & ANDREW PARTRIDGE, GENDERED NATURE OF DIGITAL INEQUALITY: EVIDENCE FOR POLICY CONSIDERATIONS 6–17 (2022), https://www.unwomen.org/sites/default/files/2022-12/BP.1_Alison%20Gillwald.pdf [https://perma.cc/Y7Z5-2D2R]; *see supra* Table 1 (documenting larger gender gaps in internet usage by women in poorer countries as compared to richer ones).

274. DAVIDE BARBIERI ET AL., EUR. INST. FOR GENDER EQUAL., GENDER EQUALITY INDEX 2020: DIGITALISATION AND THE FUTURE OF WORK 19–29, 68–111 (2020), https://eige.europa.eu/sites/default/files/documents/mhaf20001enn_002.pdf [https://perma.cc/G42L-X63C].

275. *See* LEGAL SERVS. CORP., *supra* note 132, at 71 (citing evidence that women comprise 74–77.4% of staff at Legal Services Corporation grantee organizations); *see also Diversity of 2022*

make up the majority of workers in "caring" and social service-oriented industries.[276] Should the trend from our survey generalize and extend to other fields, the natural assimilation of these advanced technologies could unintentionally exclude specific groups, along with the industries and clientele they predominantly support. Therefore, increased focus on addressing this issue could have benefits beyond the legal aid sector.

Our survey also showed that cost considerations are important. 62.5% of pilot participants reported that they would continue to use paid AI tools for work (Table 2), but numerous respondents said they would only continue using the tools if their organizations could afford them. Among the tested tools, pilot participants favored the lowest-priced paid tool, ChatGPT-4 (Figure 3).

While we recognize there are organizations that are already working to promote equitable access to generative AI technology through complimentary access, funding, and outreach to women, more can be done, by AI tool providers, legal aid funders, and others.

1. Equitable Access

We recommend that AI tool providers follow the lead of the companies in our pilot and others and make commitments to ensure AI tools get into the hands of legal aid nonprofits, for example, through equitable access programs. One way is to establish a pro bono program similar to "Everlaw for Good," which provides nonprofits and legal aid providers with complimentary access to Everlaw, an eDiscovery platform with a beta release of an AI assistant.[277] Another is by offering discounts off standard rates to qualifying nonprofits, which

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California Licensed Attorneys, STATE BAR CAL. (2022), https://publications.calbar.ca.gov/2022-diversity-report-card/diversity-2022-california-licensed-attorneys [https://perma.cc/Z5RX-SDQ4] (stating that women comprise only 44% of lawyers in California).

276. *Gender Divide in Healthcare Professions*, ROSALIND FRANKLIN UNIV. MED. & SCI., https://www.rosalindfranklin.edu/symposiums/wish/gender-divide/ [https://perma.cc/FTB6-QD4T]. For example, in healthcare and teaching, where women comprise over 75% of workers. *Id.* (though stating that women comprise only 34% of practicing physicians); Michael Hansen & Diana Quintero, *How Gender Diversity Among the Teacher Workforce Affects Student Learning*, BROOKINGS (July 10, 2018), https://www.brookings.edu/articles/how-gender-diversity-among-the-teacher-workforce-affects-student-learning/ [https://perma.cc/SB29-AXTJ].

277. *Everlaw for Good*, EVERLAW, https://www.everlaw.com/everlaw-for-good/ [https://perma.cc/UPA5-FZ49]. *See also Justice for Change,* RELATIVITY, https://www.relativity.com/company/commitments/social-impact/justice-for-change/ [https://perma.cc/D5XT-PQHR].

Gavel.io, OpenAI, and Co-Counsel have each done.[278] Other AI vendors are including legal aid groups in their beta programs. For example, Casetext worked with several legal aid groups, including the Innocence Project, during beta testing of CoCounsel.[279] It would be worthwhile to consider establishment of a norm across the legal tech industry of making AI and automation services accessible to nonprofits.

2. Funding

Legal aid organizations need more funding to procure and implement AI solutions. LSC recently awarded federally funded Technology Initiative Grants ("TIG") to legal services organizations in Los Angeles, Kansas, and Tennessee to support the use of AI.[280] But TIG funding is available only to LSC-funded legal services organizations, and more government or private funding is needed to support legal services organizations' adoption of AI.

3. Outreach

We recommend increased outreach to encourage legal aid providers, particularly women, to explore the use of generative AI. As discussed above, the State Bar of California Office of Access and Inclusion partnered with us to invite legal aid lawyers to pilot AI tools. Similar pilots and outreach could be facilitated through networks of

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278. In October 2024, CoCounsel announced that it was launching an AI for Justice Legal Aid program that includes, as one its prongs, "access to CoCounsel at a steeply reduced price." *See CoCounsel*, *supra* note 117. OpenAI offers a 20-50% discount on subscriptions for qualifying nonprofits. *OpenAI for Nonprofits*, OPENAI, https://help.openai.com/en/articles/9359041-openai-for-nonprofits (last visited Dec. 20, 2024) [https://perma.cc/2VNU-23GZ]. Gavel.io, as described above, at the time of the trial, also offered discounts to non-profits, students, and education institutions. *Bar Association Partnerships*, GAVEL (Mar. 20, 2024), https://www.gavel.io/resources/bar-associations [https://perma.cc/XH8B-9YX2].

279. *See, e.g.*, Sarah Martinson, *How Legal Aid Groups Are Using Artificial Intelligence Tool*, LAW360 (May 30, 2023, 7:02 PM), https://www.law360.com/pulse/articles/1585913 [https://perma.cc/MA27-FELX].

280. *See, e.g.*, Press Release, Legal Servs. Corp., *supra* note 50 (describing $233,210 grant to Legal Aid Foundation of Los Angeles to "develop AI capabilities within its LegalServer case management system . . . to use AI to help analyze case data to generate summaries, identify additional legal needs, and recommend appropriate case outcomes"; $314,140 to Kansas Legal Services to enhance a prior website upgrade that had boosted KLS's traffic by over 25% by "leveraging AI technology to assess and simplify legal information"; and $329,027 to West Tennessee Legal Services to leverage AI in connection with the development of a centralized data repository and predictive analytics model).

legal aid professionals, such as LSC and the Legal Aid Association of California.

In summary, no matter how useful AI tools can be, simply making them "available" will not necessarily result in their equitable uptake and application, especially if the current disparities in use, by gender or otherwise, are not directly addressed. More proactive efforts may be needed to ensure that access to and usage of the technologies is optimized within the legal aid sectors and does not tilt along traditional gender and resource lines.

B. Enhancing the Use of Generative AI in Legal Aid Through Innovative Models of Assistance

As discussed, another finding of our pilot was that various forms of assistance and information sharing enhanced the benefits of AI tools as to some outcomes. The differences were not large, but they were significant on a number of metrics. Yet few legal aid organizations can afford to dedicate resources just to supporting the use of generative AI tools. Based on our experience with this pilot, we recommend exploring the following ideas to provide assistance to legal aid professionals:

1. National Legal Aid Tech Help Desk

One way to provide concierge-type assistance at scale is to implement something like a National Legal Aid Tech Helpdesk: it would provide assistance to legal aid organizations and professionals as they learn to use generative AI and other while mitigating their risks. Particularly for organizations that are not large enough to have a large IT staff, this could provide an important infrastructural element.

2. Community of Practice

Organizations serving legal aid professionals can cultivate a community of practice for discussing and disseminating best practices on the use of AI. For example, LSC[281] and Legal Services National Technology Assistance Project[282] have facilitated discussion of legal

281. *See, e.g.*, *Technology*, LEGAL SERVS. CORP., https://www.lsc.gov/i-am-grantee/model-practices-innovations/technology [http://perma.cc/7ZTY-892Z]; *LSC's Innovations in Technology Conference*, LEGAL SERVS. CORP., https://www.lsc.gov/events/events/lscs-innovations-technology-conference [https://perma.cc/847K-JKUG]; *Talk Justice: An LSC Podcast*, LEGAL TALK NETWORK, https://legaltalknetwork.com/podcasts/talk-justice/ [https://perma.cc/57LC-XR95].

282. *About*, LEGAL SERV. NAT'L TECH. ASSISTANCE PROJECT, https://www.lsntap.org/node/523/about [https://perma.cc/LT47-SVRY]. Legal Services National Technology Assistance

technology and information sharing through conferences, podcasts, and online resources.

3. Tech Bono

To facilitate the flow of information, a new "Tech Bono" model could be adopted, where tech-savvy professionals or law students volunteer in legal clinics or pro bono matters, leveraging generative AI and sharing their methods with clients and counsel. As discussed below, one of us did so during the pilot with positive results. An extension of the pro bono model would be for engineers and law students to work together with organizations to serve not only individual clients, but clients at scale through automation and AI tools.

Such support structures may assist legal aid organizations in harnessing the potential of generative AI and also promote a culture of knowledge sharing and collaboration.

C. Facilitating Tech + Legal Aid Lawyer, not Tech v. Legal Aid Lawyer Offerings

The beneficial use cases submitted by our pilot participants generally focused on lower-risk tasks where generative AI's role was to augment and support, rather than replace, legal aid lawyers. This included helping lawyers get to a good first draft, edit a draft, summarize documents, or perform a variety of nonlegal tasks (e.g., draft a grant application or update a website). While pilot participants reported time savings of up to 80% on certain tasks, they also emphasized the need to verify, correct, and/or supplement AI-generated outputs.[283] The outlook of pilot participants on the impact of AI technologies on their jobs was generally favorable: 71% reported that they believed AI would replace elements of their work in a positive way. (Table 2) Another 26% of respondents expressed that they did not expect AI to replace any element of their jobs, while only 3% of respondents selected

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Project (LSNTAP), a part of the Michigan Advocacy Program, helps legal aid organizations effectively use technology by providing training, information, online tools, and community forums. *Id.* Among other things, LSNTAP has created a database of cases, attorney guidance, judicial opinions, and state bar materials on the use of generative AI. *See AI & Legal Information Database*, LEGAL SERV. NAT'L TECH. ASSISTANCE PROJECT, https://www.lsntap.org/node/662/ai-legal-information-database [https://perma.cc/YSP5-TS7A].

283. *See infra* Appendix, Table A2.

the option associated with a belief that AI's impact on their work would be negative.[284]

The synergy between the role of generative AI and the human lawyer became clear for one of us while volunteering in legal aid clinics under the supervision of two of our pilot participants from the Lawyers' Committee for Civil Rights of the San Francisco Bay Area (LCCR). During the first 40 minutes or so of a 60-minute clinic appointment, the participating pro bono and LCCR attorneys would ask questions about the client's legal needs. During these discussions, the lawyers' empathy and judgment were critical to understanding the often sensitive needs of the client. After obtaining consent from the client and all participating attorneys, generative AI tools (e.g., ChatGPT-4, Bard, Bing) were used to address the client's needs by drafting a letter (e.g., to a landlord or government agency) and customizing the letter for review and approval by the supervising LCCR attorney. This enabled the LCCR attorney to provide the client with a customized letter by the end of the appointment, resulting in faster and better work product for clinic clients.

As was the case in the LCCR clinic, the integration of generative AI into legal aid has the potential to enhance the capabilities of legal aid lawyers, enabling them to serve clients more efficiently and effectively. We recommend that legal aid see generative AI as a powerful assistant to lawyers, not as a replacement, ensuring that legal services become more accessible while retaining the essential human expertise and empathy that is central to legal aid.

D. Facilitating the Development of Legal Aid-Directed Technological Solutions

While getting technology in the hands of legal aid lawyers is important, it will not fully close the justice gap. 92% of low-income Americans do not receive any or adequate help for their civil legal problems,[285] and the ABA has estimated there are only 2.8 paid legal aid lawyers for every 10,000 residents in poverty.[286]

Therefore, we recommend that tool vendors and legal aid professionals work together to develop more AI-powered solutions,

284. *See supra* note 161.

285. LEGAL SERVS. CORP., *supra* note 20, at 7.

286. AM. BAR ASS'N, PROFILE OF THE LEGAL PROFESSION 7 (2023), https://www.americanbar.org/content/dam/aba/administrative/news/2023/potlp-2023.pdf [https://perma.cc/Y8ZA-7VXC].

including consumer-facing solutions, not just for legal services organizations and the individuals whom they serve, but also more generally for low-income consumers. Legal aid-directed solutions that do not require interaction between the legal aid workforce and those utilizing the solutions may hold the greatest potential to use generative AI to help close the justice gap. Legal aid lawyers are at the forefront of the justice gap, so they are well-qualified to help design solutions that will increase their efficiency, reach, and impact.

Based on the current limitations of generative AI and the feedback from our pilot participants, we believe that combining generative AI capacities with automation technology and human oversight can lead to the creation of effective, scalable consumer-facing solutions. Here are a few examples of such solutions developed by lawyers in three areas of legal aid assistance—eviction defense/housing, criminal expungement, and immigration—where the ability of the consumer to interact directly with the technology appears not just marginally, but dramatically, to increase service coverage. These solutions can serve as inspiration for further discussion, research, and development of legal aid-directed technologies.

1. Housing

A joint effort of Law Center for Better Housing ("LCBH") and the Lawyers Trust Fund of Illinois, Rentervention is a generative AI platform that helps Illinois tenants with housing issues.[287] When tenants visit Rentervention.com, they can chat with Renny, a virtual AI assistant powered by OpenAI's GPT-4. Renny helps clients diagnose their legal issue and provides relevant, human-verified legal information about their needs. If Renny determines that LCBH can assist with the tenant's legal issue, the tenant is directed to an automated workflow built using Gavel's no-code software platform. Renny can then help the tenant draft an automated letter to their landlord. If necessary, Renny can connect the tenant to a staff member who will follow up in one business day.[288] By combining legal technology and

287. RENTERVENTION, www.rentervention.com [https://perma.cc/95VN-CXX5].

288. *Id.; see also About*, RENTERVENTION, https://rentervention.com/about/ [https://perma.cc/PH5C-ZJVC] (explaining that tenants can use Renny for legal advice regarding housing issues); Gavel, *How Rentervention Used OpenAI/GPT and Gavel to Provide Legal Services*, YOUTUBE (Jan. 26, 2024), https://www.youtube.com/watch?v=pLH66GJq5KI [https://perma.cc/DW6Y-9GCM]; Connor Malloy, *Gavel and GPT: Empowering Tenants with Rentervention*, LOOM (Jan. 2024), https://www.loom.com/share/358284de65e7475cb4dde623e3ee32cf?sid=0a30a501-31da-4

human support, "Rentervention has helped more than 50,000 renters since 2019,"[289] likely many times more than how many would be served using a standard clinic model.[290]

2. Expungement

Rasa Legal, created by former public defense lawyer Noella Sudbury, shows the promise of legal aid-lawyer-driven technology offerings for a more complex legal service: expungement. As a for-benefit corporation, Rasa uses technology to streamline the expungement process, which, due to its complexity, means that an estimated 75% of self-represented individuals fail to complete the expungement process.[291] But using technology to allow people to search for and view their own records,[292] determine expungement eligibility, and file the necessary paperwork, Rasa has been able to achieve much greater efficiency and service delivery.[293] According to Sudbury, in Rasa's first year, "close to 10,000 people used [their] services (in comparison to 400 people per year served by legal aid)."[294]

3. Immigration

The American Immigration Lawyers Association (AILA), a national 16,000-member association, collaborated with Visalaw.Ai to develop GEN 1.0, a generative AI-powered legal research product designed to assist immigration lawyers. Launched in January 2024, GEN

93f-a4cc-4df98cb91893 [https://perma.cc/GBT7-DMP3]. Renny is also available within LCBH to help staff members navigate LCHB's legal knowledge base and prepare responses to tenants' questions. Conor Malloy, *Update on Rentervention Helpdesk*, LOOM (Mar. 1, 2024), https://www.loom.com/share/6b2dc5c8e9c74e1a9c92b0bffd613fda [https://perma.cc/49KR-66XM].

289. *Rentervention Named Winner at 22nd Annual Chicago Innovation Awards*, LAW CTR. FOR BETTER HOUS. (Nov. 17, 2023), https://lcbh.org/rentervention-named-winner-at-22nd-annual-chicago-innovation-awards/ [http://perma.cc/U24V-5W2Z].

290. For example, at the same time that Rentervention was recently honored, a law school clinical project facilitated volunteer work on twenty eviction defense cases, some of them quite complex, over the same period. *Law Center for Better Housing Honors Professors Laurie Mikva and Daniel Linna*, NW. PRITZKER SCH. LAW (Sept. 21, 2023), https://news.law.northwestern.edu/law-center-for-better-housing-honors-professors-laurie-mikva-and-daniel-linna/ [https://perma.cc/HZ3Z-894W].

291. *The Reason for Rasa*, RASA, https://www.rasa-legal.com/about/the-reason-for-rasa/ [https://perma.cc/PM3W-LFD2].

292. Robby Poffenberg & Mark Tullis, *Rasa Legal Uses Tech to Clear Criminal Records and Create Opportunity*, TECHBUZZ (Mar. 28, 2023), https://techbuzz.news/rasa-legal-uses-tech-to-clear-criminal-records-and-create-opportunity/ [perma.cc/R7BB-4DMJ].

293. *Id*.

294. Email from Colleen Chien to Noella Sudbury (Sept. 21, 2023, 4:28 PM) (on file with author).

is built on GPT-4, and its knowledge base includes an immigration law library and AILA resources that are not publicly available.[295] An earlier version was beta-tested by members of the AILA, demonstrating the broad impact that tools can have in a short period of time.[296] On its website, Visalaw.Ai features a variety of use cases such as legal research, eligibility analysis, and summarizing documents.[297]

E. Certification for Legal Aid Bots

Some of the most promising use cases that we heard about involved chatbots that can converse with consumers and dispense relevant legal information in many different forms and languages and customize the information in accordance with the user's specific needs. However, going from a concept to a reliable and deployable bot solution involves a high level of vetting and quality control, which would benefit from updates and adaptation of the regulation that is done by human lawyers, paraprofessionals, or non-lawyer community-based advocates. Therefore, we recommend that the legal industry consider exploring regulatory sandboxes or community-based justice worker models that involve standards and tech certifications to help ensure the deployment of high-quality, trustworthy AI tools for legal aid.

To protect the public interest, regulatory entities generally restrict the practice of law to legal professionals who have passed the bar exam and character and fitness requirements, maintain ethical standards, fulfill continuing legal education requirements, and uphold their duties to their clients and the public.[298] As discussed above, Utah has implemented a regulatory sandbox to allow for the development and testing of innovative legal services models.[299] Utah, Arizona, and several other states have developed community-based justice worker models that allow non-lawyers to be trained and certified to provide

295. Press Release, Am. Immigr. Law. Ass'n, Visalaw.Ai Launches GEN 1.0: A Revolutionary AI-Powered Legal Research Tool for Immigration Law Firms (Jan. 16, 2024), https://www.aila.org/library/visalaw-ai-launches-gen-1-0-a-revolutionary-ai-powered-legal-research-tool-for-immigration-law-firms [https://perma.cc/2RYJ-WVDH]; Josh Sinnott, *Where Does GEN's Knowledge Come From?*, VISALAW.AI (Sept. 28, 2023), https://www.visalaw.ai/blog/gen-knowledge [https://perma.cc/GSU9-MFUJ].

296. Andrew Kreighbaum, *AI Makes Its Way to Immigration with New Tool to Aid Attorneys*, BLOOMBERG L. (June 21, 2023, 7:21 AM), https://news.bloomberglaw.com/daily-labor-report/ai-makes-its-way-to-immigration-with-new-tool-to-aid-attorneys [https://perma.cc/3F94-R4P6].

297. *Use Cases*, VISALAW.AI, https://www.visalaw.ai/usecases [http://perma.cc/9VLT-F8UZ].

298. MODEL RULES OF PRO. CONDUCT r. 5.5 (AM. BAR ASS'N 2019).

299. *See* discussion *supra* Section I.B.

limited-scope legal advice in certain areas, including housing, family law, and medical-legal debt.[300] These regulatory reforms could be explored by other jurisdictions to cover new models that are intended to directly serve the public.

In addition, state bar authorities could consider whether to institute and oversee voluntary certification or "seal of approval" programs verifying that automated legal services meet appropriate functionality, accuracy, and ethical standards. Particularly in light of the known challenges of generative AI technology including bias, accuracy, privacy, and confidentiality,[301] a certification or "seal of approval" program could evaluate legal bots across dimensions including reliability in providing accurate and up-to-date legal information, ability to spot ethical issues, transparency around limitations and disclaimers, and adherence to privacy policies and data regulations. Lawyers and technologists could collaborate on appropriate certification standards and procedures tailored for verifying legal bots across domains and dimensions.

CONCLUSION

In conclusion, the integration of generative AI into legal aid presents an opportunity to enhance access to justice. However, our surveys and pilot findings highlight the need for thoughtful and strategic approaches to maximize access and effective utilization of generative AI, as well as the importance of addressing the gender gap in its use. We encourage further collaboration and dialogue among the legal aid community, technology providers, researchers, and policymakers with a focus on the development of legal aid-directed solutions that combine generative AI and automation technology with proper human oversight and quality controls.

300. *Id.*
301. *See* discussion *supra* Section IV.A.

APPENDIX

- Methodological Details, Demographics of Survey Takers, and Tables
- Survey 1 & 2 Instruments

METHODOLOGICAL DETAILS

Data from the initial study and the pilot survey were combined in order to examine baseline and post-pilot differences between respondents. Word responses were normalized to numerical responses for purposes of display and analysis, according to the scales described in Part III. As items were assessed on different response scales, some of the graphic representation of results include responses normalized to a 1–5 point scale.

Descriptive statistics including means, standard deviations, frequencies, and percentages were computed for each variable of interest. We then assessed mean differences in response items between different subgroups (i.e., gender, age group, role, experience level) using independent sample t-tests or ANOVA. We assessed proportion differences in categorical response items between subgroups with chi-square tests or Fisher's exact tests, when the estimated cell count was less than 5. To verify the robustness of our results, we calculated adjusted means and proportions by estimating regression models including covariates such as current/previous AI use.

DEMOGRAPHICS OF SURVEY TAKERS

Out of the 204 respondents to the initial survey, 88.1% were attorneys and 11.9% were non-attorneys. Among individuals that completed the pilot (P1), the ratio was 95.2% attorney and 4.8% non-attorney. Our responses skewed female, consistent with the demographics of legal aid practitioners[302]—among people for whom

302. LEGAL SERVS. CORP., *supra* note 133, at 71. 73.8% of all full-time staff and 77.4% of part-time staff at Legal Services Corporation grantees were women in 2022. *Id.* In comparison, men outnumber women in the law in California. Diversity of 2022 California Licensed Attorneys, THE STATE BAR OF CALIFORNIA, Fig. 3 (2022), https://publications.calbar.ca.gov/2022-diversity-report-card/diversity-2022-california-licensed-attorneys (44% of lawyers in California identified as female in 2022); *see also* Catherine Albiston et al., *supra* note 133, at 264 (finding that "identifying as female is predictive of public interest work at later stages of one's career but not at the beginning . . . gender *is* significantly related to holding a *current* job in a public interest setting"); *id.* at 239 n.80 (summarizing past research finding that women law students prefer to pursue public interest work).

we had binary gender data, 77.9% of the respondents to our baseline survey were women, and 22.1% were men; in the pilot study, 68.9% were women and 31.2% were men.

In terms of age, our population also skewed towards older, more experienced lawyers: 38.4% of the initial respondents were aged 45 or older, 49.6% were between 35 and 44 years old, and 11.6% were 34 or younger. Of the initial respondents, 23.4% had between 0–4 years of experience, 36.3% had 5–14 years of experience, 26.6% had 15–25 years of experience, and 13.7% had over 25 years of experience.

But our pilot study differed in important ways from our survey population. It skewed towards younger, relatively less experienced lawyers: 25.8% were aged 45 or older, 54.8% were between 35 and 44 years old, and 19.4% were 34 or younger. In the pilot study, 41.5% had 0–4 years of experience, 30.8% had 5–14 years of experience, 16.9% had 15–25 years of experience, and 10.8% had over 25 years of experience.

Of the initial respondents, 52.5% were white, and 47.6% were people of color. In the pilot study, 56.3% were white, and 43.8% were people of color. Of the initial respondents, 61.4% were in leadership, 8.4% were supervisors, 12.4% were in direct service roles, and 17.8% were in other roles. In the pilot study, 47.6% were in leadership, 7.9% were supervisors, 19.1% were in direct service roles, and 25.4% were in other roles.

Table A1: Differences in outcomes by gender in the pilot population before and after the pilot[303]

PRE-PILOT	Men	Women
Currently Using	65%	22%*
Number ways used	2.1 (1.8)	0.8 (1.7)*
Number ways considered use	1.9 (1.5)	2.9 (2.4)**
Benefit	3.6 (0.8)	3.1 (0.9)+
Number of Concerns	3.1 (1.7)	3.8 (1.8)

POST-PILOT	Men	Women
Used How Often	3.3 (0.8)	3.1 (1.1)
Usage during pilot compared to before (adjusted for previous use)	2.9 (0.7)	2.8 (0.6)
Number ways used	4.7 (1.9)	3.8 (1.6)
Quality of AI output	1.6 (0.5)	1.8 (0.5)
Comparative Satisfaction	2.7 (0.5)	2.7 (0.5)
Productivity Increase	90%	90%
Level of Concern about AI	79%	63%
Number of Concerns	5.1 (1.9)	4.5 (2.1)
Positive impact of AI on my job	63%	78%
Future Use (any)	74%	76%
Yes Future Paid Use (any)	58%	63%

Note: + $p < .10$, * $p < .05$, ** $p < .01$

303. Calculated on the basis of the 66 individuals' responses to surveys before and after the pilot. Because some of the respondents to the initial survey received a slightly different version of the questions pertaining to number of ways used, considered, and concerns, as described *infra* at Appendix Survey 1, the pre and post-pilot absolute values are not directly comparable.

Table A2: Differences in outcomes by concierge or standard support during the pilot

	Unadjusted		Adjusted for Previous Use	
POST-PILOT	Standard Group (N=41)	Concierge Group (N=25)	Standard (N=41)	Concierge Group (N=25)
Used How Often	3.0 (1.0)	3.3 (1.0)	3.1	3.3
Usage during pilot compared to before	2.8 (0.5)	2.7 (0.7)	2.8	2.7
Number ways used	3.8 (1.9)	4.3 (1.7)	3.9	4.2
Quality of AI output	1.6 (0.5)	1.9 (0.4)*	1.6	1.9**
Comparative Satisfaction	2.6 (0.6)	2.8 (0.4)+	2.6	2.8
Productivity Increase (any improvement)	84%	100%*	87%	99%
Productivity Increase (1-4 scale)	2.2 (0.8)	2.8 (0.8)**	2.2	2.9**
Level of Concern about AI	73%	67%	74%	65%
Number of Concerns	4.5 (2.1)	4.9 (2.0)	4.5	4.9
Positive Impact of AI on my job	62%	88%*	61%	88%*
Future Use (any)	73%	79%	74%	77%
Yes Future Paid Use (any)	55%	75%+	55%	74%

Note: + $p < .10$, * $p < .05$, ** $p < .01$. Adjusted means and percentages were obtained by running multiple regression models predicting each item from group type (concierge vs standard) and adding previous use (yes/no) as a covariate.

SURVEY 1 (ACCESS TO JUSTICE AI AUTOMATION + PILOT SURVEY COMBINED—USING FIRST SURVEY AS BASE)

Hello, we are researchers at UC Berkeley studying access to justice and AI/automation tools. We invite you to take the following anonymous survey, which should take no more than 10 minutes. We will use your responses to write one or more research articles about the potential of AI and automation to help address the justice gap but will not identify you or your organization in our research. Your participation is voluntary, and you are free to opt out at any time without penalty. We will provide a $10 gift card for your completion of the survey. You may forward this to legal services providers who directly serve clients at your organization, but each person is only eligible to receive one gift card. At the end of the survey, you'll have the opportunity to sign up for our pilot project which will give participants free trial access to paid GenerativeAI platforms in exchange for providing a written or interview-based debrief about the experience.

This survey is being implemented pursuant to University of California Berkeley IRB Protocol 2023-10-16780. You are welcome to email Miriam at miriam.kim@berkeley.edu or University of California Berkeley's Office for Protection of Human Subjects (OPHS) at ophs@berkeley.edu or 510-642-7461 if you have questions or concerns about the survey.

By answering the questions below, you consent to participating in research.[304]

Thank you for your participation,

Prof. Colleen Chien and Miriam Kim, Berkeley Law

<u>Gating and Background</u>

1. Have you previously taken a survey from the A2J/AI team at Berkeley?[305]
    a. Yes
    b. No

304. A slightly different version of this appeared in the second distribution of the survey.
305. This question (1) appeared only on the second distribution of the survey, and not the first.

2. Please provide your name, contact, information, and organization.[306]

3. Please provide the names and contact information of other attorneys at your organization you'd like us to reach out to offer the pilot to (optional) You may also share with them this link: https://bit.ly/A2JAIPilot[307]

4. Are you a lawyer?[308]
    a. Yes
    b. No
5. In which of the following areas do you practice? Select all that apply.
    a. Children / youth
    b. Civil rights
    c. Conservatorship
    d. Consumer / economic justice
    e. Criminal
    f. Disability rights
    g. Domestic violence
    h. Education
    i. Elder law
    j. Employment / workers' rights
    k. Family law
    l. Guardianship
    m. Health
    n. Housing
    o. Immigration
    p. Income maintenance
    q. Juvenile
    r. Litigation
    s. Miscellaneous
    t. Veterans' rights
    u. Voting rights

306. This question (2) appeared only on the second distribution of the survey, and not the first.
307. This question (3) appeared only on the second distribution of the survey, and not the first.
308. This question (4) was worded differently in the second distribution, instead asking "are you employed at an organization that provides legal services to or advocates on behalf of underserved populations, directly or indirectly?"

v. Women's rights
w. Other _____

6. Please identify the three tasks on which you spend the most time:
    a. Client intake
    b. Community clinics
    c. Community education
    d. Community outreach
    e. Direct legal services
    f. Litigation
    g. Marketing/communications (e.g., blog posts, website, social media)
    h. Policy work
    i. Development (e.g., fundraising, grant writing)
    j. Recruiting staff and volunteers
    k. Administrative tasks
    l. Operations
    m. Other ____

7. Please indicate your role in the organization for which you work (check all that apply)
    a. Leadership
    b. Supervisor of attorneys /volunteers to provide direct services to clients
    c. Provide direct services to clients
    d. Other [let them choose]

<u>Usage of Legal Tech and AI</u>

8. Are you using or do you plan to use Generative AI (ChatGPT, Claude, Bard, Co-Counsel, etc.) tools in your work?
    a. Currently using
    b. Planning to use
    c. Not using and do not plan to use
    d. Not sure if I will use

If *"a: currently using"* is not selected, start Perceptions of AI:

Perceptions of AI

8a. Would you consider using AI tools for the following tasks at work? Select all that apply.[309]
    a. Researching matters
    b. Increasing efficiency
    c. Writing emails
    d. Drafting documents
    e. Understanding new legal concepts
    f. Streamlining work
    g. Improving work quality
    h. Document analysis
    i. Conducting due diligence
    j. Translation
    k. Other
    l. Developing litigation strategies

8b. What concerns do you have about using AI tools in your work? (select all that apply)[310]
    a. No concerns
    b. Bias
    c. Confidentiality
    d. Cost
    e. Data Privacy
    f. Ethical concerns
    g. Hallucinations
    h. Inability to explain how AI works
    i. Inaccurate results
    j. Other _____

End "Perceptions of AI ARM"
If *"a: currently using"* is selected, then start "AI Tech Arm:"

309. This question (8a) in the second distribution of this survey included the instruction "Select the top 3 areas of interest," rather than "Select all that apply."

310. This question (8b) in the second distribution of this survey included the instruction "Select your top 3 answers," rather than "Select all that apply."

AI Tech Arm

8c. How are you using Generative AI tools in your work? Select all that apply.[311]
   a. Researching matters
   b. Increasing efficiency
   c. Writing emails
   d. Drafting documents
   e. Understanding new legal concepts
   f. Streamlining work
   g. Improving work quality
   h. Document analysis
   i. Conducting due diligence
   j. Translation
   k. Other
   l. Developing litigation strategies

8d. How have these tools helped or hindered your practice? Please describe the tools you are using and provide examples regarding how you are using them.

8e. Please indicate if you'd be interested in talking to us further about your experiences with Generative AI.[312]
   a. Yes
   b. No

8f. What concerns do you have about using AI tools in your work? (select all that apply)[313]
   a. No concerns
   b. Bias
   c. Confidentiality
   d. Cost
   e. Data Privacy
   f. Ethical concerns

311. This question (8c) in the second distribution of this survey included the instruction "Select your top 3 answers," rather than "Select all that apply."

312. This question (8e) only appeared in the first distribution of the survey and was later edited out for the second distribution.

313. This question (8f) included "select your top 3 answers," rather than "select all that apply" in the second distribution of the survey.

g. Hallucinations
h. Inability to explain how AI works
i. Inaccurate results
j. Other: _____

End "AI TECH ARM"

9. Based on what you know about AI conversational tools like ChatGPT, how beneficial do you believe they are in assisting with tasks or answering questions?[314]
   a. Very Beneficial
   b. Beneficial
   c. Somewhat beneficial
   d. Neither beneficial nor harmful
   e. Somewhat harmful
   f. Harmful
   g. Very harmful

10. Are you using primarily paper-based records for any of the following tasks? Select all that apply.[315]
    a. Client intake
    b. Case management
    c. Legal research
    d. Legal forms and petitions
    e. Timesheets/payroll
    f. Other: ____
    g. None of the above

11. Which of the following limits the number of people served by your organization? (select all that apply)[316]
    a. Knowledge gap: people don't know they have a legal remedy
    b. Eligibility: people who come to us often are ineligible
    c. Inaccessible court records or government information

314. This question (9) only appeared in the second distribution of the survey.

315. This question (10) appeared differently in the second distribution of the survey, instead stating "If you are using primarily paper-based records for any of the following tasks, please indicate that below."

316. This question (11) included "select the top 3 answers," rather than "select all that apply" in the second distribution of the survey.

d. Marketing: people don't know about us or our services
e. Not enough funding
f. Not enough legal aid lawyers
g. Not enough time
h. Translation issues
i. Unauthorized practice of law (UPL) regulations
j. Other

12. Please expand, if you'd like (optional).

13. If asked by a funder how you could double your impact, in terms of clients served, what resources would you ask for and why?[317]

14. Please indicate if you would like to:
    a. receive a gift card for your time.
    b. be contacted for future paid survey or interview opportunities, or to elaborate on your response.

15. Please provide your contact information (name, email, and organization name) so that we may follow up.

16. We have a limited number of free subscriptions to GenAI tools like ChatGPT-that we are offering to direct service providers as part of a pilot. The attorneys would get free 1-month subscriptions in exchange for sharing their experiences through a brief interview or short report (for which a $50 gift card will be offered). Would your organization like to participate? Please only check "yes" / "maybe" if your organization does not prohibit the use of such tools.
    a. Yes
    b. No

317. This question (13) marks the end of the second distribution survey. Questions 14 to 16 appeared only in the first distribution.

17. If you would like to participate in our pilot and receive free subscriptions to AI tools for a month, please indicate that by selecting the option below:[318]
    a. Yes, I consent to participating in the pilot and have a ChatGPT account already ready to be upgraded
    b. Yes, I consent to participating in the pilot but need to set up a ChatGPT account.
    c. No, I cannot participate in the pilot at this time, but please contact me again.
    d. No, I'm no longer interested but please send me a gift card for finishing the survey.
    e. No, I'm no longer interested and do not want a gift card for finishing the survey.

318. This question (17) appeared only on the second distribution of the survey was placed at the front.

SURVEY 2 (PILOT EXIT SURVEYS FOR CONCIERGE + STANDARD GROUP)

Thank you for participating in the Berkeley Law pilot. To wrap up, we are asking for feedback in two parts: your responses to this survey and your documentation of 2–5 use cases. The survey should take no more than 5 minutes and the use cases should take 1–5 minutes per example to document (Instructions: https://bit.ly/AIA2JShare). Do not include any confidential information or PII; please modify or redact your use cases to anonymize them, if necessary.

We will provide you a $50 gift card if you complete the survey and report on or before January 12, 2024, to be sent in batches the weeks of January 1 and 15.

If you want to be acknowledged for your submissions, please indicate that in your use case entry, and we will provide you with credit as you wish. We will not identify you or your organization without your consent in connection.

Your participation is voluntary and you are free to opt out at any time without penalty. This survey is being implemented pursuant to University of California Berkeley IRB Protocol 2023-10-16780. Please email Miriam at miriam.kim@berkeley.edu or University of California Berkeley's Office for Protection of Human Subjects (OPHS) at ophs@berkeley.edu or 510-642-7461 if you have questions or concerns about the survey.

Thank you for your participation in our pilot and we hope you will continue to explore the use of AI to increase access to justice.

- Prof. Colleen Chien and Miriam Kim, Berkeley Law

Identify Yourself

1. What is your name?

2. What is your email?

Which tools were used

3. Which tools had you used prior to the pilot, if any?
    a. ChatGPT Plus
    b. ChatGPT
    c. Claude
    d. Bard

e. Bing Chat
f. CoCounsel
g. Gavel

4. How did your use of the following tools change during the pilot, if at all? (0= NA, not available, 1= no change, 2= a bit more, 3= a lot more) as applied to
    a. ChatGPT Plus, ChatGPT, Claude, Bard, Bing Chat, CoCounsel, Gavel

For Concierge service group[319]

5. Which of the following **services** did you receive as part of the pilot, and if you used them, please indicate how useful they were:[320] (0 = did not receive, 1= received but did not use or find useful, 2= received and found pretty useful, 3= received and found very useful)
    a. Google group
    b. Tool Trainings
    c. Weekly best practices emails
    d. Office hours
    e. Emails
    f. FAQs

6. Please comment, if you wish (optional)

7. How could we have done better? (optional)

For all (both standard and concierge group)

8. What outside sources of information did you find useful for using the AI tools? (select all that apply)
    a. Tool websites
    b. Peers

319. This section (Q5 to Q7) was only shown to the concierge group, and not the standard group.

320. For FAQs, see *ChatGPT/Claude Pilot FAQ's*, https://docs.google.com/document/d/1ozYhO0vUbENKSScMnMv-nHZVDzVTMl6xyA0I4cBHNfg/edit#heading=h.r9aj24v4gg4p [https://perma.cc/C2N6-VTTZ].

c. Internet search
d. Facebook or other opt-in group (e.g. slack or mailing list)
e. Other _____

Nature and quality of the experience

9. What was the extent of your usage of generative AI during the pilot?
    a. Everyday
    b. Most days
    c. Some days
    d. Occasionally
    e. None

10. Compared to before the pilot, your usage of Generative AI tools during the pilot was
    a. More than before
    b. The same as before
    c. Less than before

11. Select all of the ways that you used the technology:[321]
    a. Brainstorming or ideation
    b. Research
    c. Legal writing
    d. Nonlegal writing
    e. Translation
    f. Document summarization
    g. Analyzing data and making forecasts,
    h. Generating /checking computer code, image recognition and generation,[322]
    i. Development work
    j. Forms automation
    k. Designing your own bot

321. Conf. Bd., *supra* note 136, at 95.
322. *Id.*

12. Please indicate the level you found your AI's output quality to be comparable to:
    a. Expert worker
    b. Experienced worker
    c. Novice worker

13. Please indicate your level of satisfaction with your AI-assisted work process,[323] as compared to before the pilot:
    a. More satisfied
    b. Equally satisfied
    c. Less satisfied

14. Did you experience an increase in productivity as a result of the AI tools? 0=NA, 1 = no increase, 2 = a little bit of an increase, 3 = a medium increase, 4 = a significant increase

<u>Attitudes towards and likelihood of continued usage</u>

15. Did the pilot make you more or less concerned about the potential use of AI tools in your work?
    a. More concerned
    b. Just as concerned
    c. Less concerned

16. Please indicate your concerns about AI:
    a. Bias
    b. Confidentiality
    c. Cost
    d. Data Privacy
    e. Ethical concerns
    f. Hallucinations
    g. Inability to explain how AI works
    h. Inaccurate results
    i. Other _____

323. Choi et al., *supra* note 94 (manuscript at 36) ("young lawyers provided with access to AI to facilitate their work accurately appreciate these benefits of AI, find that access to AI tends to enhance their work satisfaction, and generally become more enthusiastic about using AI to facilitate their work as they gain experience doing so.").

17. Please check the box that best describes your outlook on the role of AI in your work:
    a. AI will replace elements of my job in a positive way—e.g., by freeing up time for more valuable or creative tasks.
    b. AI will replace parts of my work in a negative way—e.g., by threatening my job altogether.
    c. I do not expect AI to replace any element of my job.

<u>Future usage</u>

18. Based on your experience, do you plan to continue to use AI tools in your work and if so, which ones?
    a. No
    b. Yes, ChatGPT Plus
    c. Yes, Claude
    d. Yes, Bard
    e. Yes, Bing
    f. Yes, Co-counsel
    g. Yes, Gavel

19. Please comment, if you wish (optional)

20. Are there features or capabilities you would like to see added or improved? (optional)

<u>Open-Ended Question</u>

21. We hope to disseminate our work to policymakers interested in how to use technology to make government systems more accessible and equitable. What technical or data access improvements could local or state governments make to make your work easier? Please be specific and explain the impact this capacity would have on expanding access. (For example, "for us, the bottleneck is accessing and reading rap sheets which requires Livescans which are costly and take time. Thus, please tell the California government to improve rap sheet access and also to give us the person's record in a more readable form.")

Demographics/Background

22. Are you a lawyer?
    a. Yes
    b. No

23. In which of the following areas do you practice? Select all that apply.
    a. Children / youth
    b. Civil rights
    c. Conservatorship
    d. Consumer / economic justice
    e. Criminal
    f. Disability rights
    g. Domestic violence
    h. Education
    i. Elder law
    j. Employment / workers' rights
    k. Family law
    l. Guardianship
    m. Health
    n. Housing
    a. Immigration
    o. Income maintenance
    p. Juvenile
    q. Litigation
    r. Miscellaneous
    s. Veterans' rights
    t. Voting rights
    u. Women's rights
    v. Other _____

24. Please indicate how you primarily self-identify your race/ethnicity:
    a. Hispanic or Latin American
    b. White
    c. Black or African American
    d. American Indian or Alaska Native
    e. Asian, Asian-American, or Pacific Islander
    f. Middle Eastern or North African

g. Multiracial
h. Other

25. Please indicate how you primarily identify:
    a. Man
    b. Woman
    c. Non-Binary
    d. Transgender Man
    e. Transgender Woman
    f. Other (e.g., gender fluid)
    g. Prefer Not to Say

26. What is your age?
    a. Under 25
    b. 25–34
    c. 35–44
    d. 45–54
    e. 55–64
    f. 65+

27. How long have you been practicing law?
    a. 0–4 years
    b. 5–14 years
    c. 15–25 years
    d. Over 25 years